

Conjugated Systems & Aromaticity

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1 Conjugated Systems

1.1 Organic Acids & Bases Review

Core Concept 1.1: General Organic Compound Acidity

Memorizing the following pK_a values as baseline references is essential—they serve as anchors for estimating the acidities of related compounds by comparing functional groups, electron-withdrawing effects, and resonance stabilization of the conjugate base.

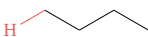
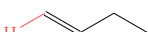
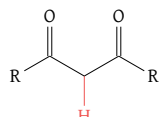
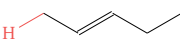
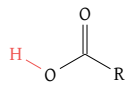
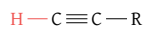
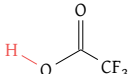
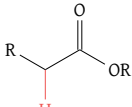
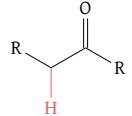
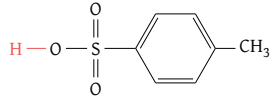
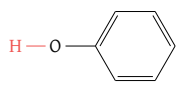
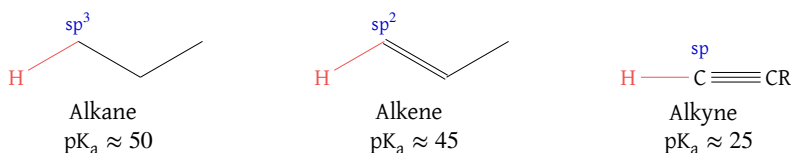
Compound	pK_a	Structure	Compound	pK_a	Structure
Alkane	50		Ammonium	10	
Alkene	45		α to Two Carbonyls	10	
Allylic	40		Carboxylic Acid	5	
Amine	35		Oxonium	0	
Alkyne	25		Trifluoro Acetic Acid	0	
α to an Ester	25		Hydrohalic Acid	<0	
α to a Ketone or Aldehyde	20		Sulfuric Acid	<0	
Alcohol (ROH)	15		Toluenesulfonic Acid	<0	
Phenol (PhOH)	10				

Table 1: Common acids and their pK_a values, ordered from least to most acidic

Core Concept 1.2: Acidity of Alkanes, Alkenes, and Alkynes

We now examine the pK_a values trend of alkanes, alkenes, and alkynes. The linear pattern observed can be rationalized by considering the following factors.

Figure 1: Alkane, alkene, and alkyne pK_a values

- (a) **Electronegativity of atom attached to the acidic H:** This factor cannot account for the trend, since the acidic hydrogen is attached to a carbon atom in all three cases, precluding any difference in electronegativity.
- (b) **Inductive Effect:** This factor is likewise insufficient, as only C–H bonds are present in the vicinity of the acidic proton.
- (c) **Resonance:** No resonance stabilization of the conjugate base is possible, so this factor does not explain the trend.
- (d) **Charge:** All species are neutral, so this factor offers no explanation.
- (e) **Hybridization:** Here a genuine difference emerges, as the acidic carbon is sp^3 -, sp^2 -, or sp -hybridized in the alkane, alkene, and alkyne, respectively. The lone pair generated upon deprotonation is stabilized most effectively in the sp -hybridized orbital, since its greater s character holds the electrons closer to the nucleus.

When the acidity cannot be rationalized from the structure of the acid itself, it is useful to instead examine the **conjugate bases**.

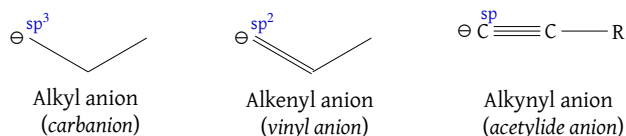


Figure 2: Conjugate bases of an alkane (left), alkene (middle), and alkyne (right)

Among the anions shown in Figure 2, the **alkynyl anion is the weakest base** because its lone pair is the most stabilized and thus least reactive. This arises because its e^- density is drawn more toward the carbon nucleus—away from the bonding region—making it less attractive toward H^+ relative to the alkenyl and alkyl anions. The following energy diagram below illustrates this comparison.

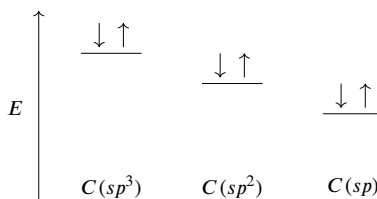


Figure 3: Diagram of sp^3 , sp^2 , and sp orbital energy levels; lone pair in the sp orbital is most stabilized because the relatively greater s character increases the probability e^- 's residing closer to the positive charge of the nucleus

Core Concept 1.3: K_{eq} and pK_a s

Any acid–base reaction has an equilibrium constant given by $K_{\text{eq}} = 10^{(\text{pK}_{a,\text{conj. acid}} - \text{pK}_{a,\text{acid}})}$. Using the pK_a values in Table 1, we can therefore estimate the K_{eq} of a given acid–base reaction to a reasonable degree of accuracy, as illustrated in the following examples.

(a) Protonation of ammonia by hydronium

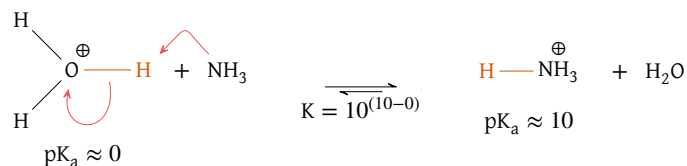


Figure 4: Protonation of ammonia by hydronium

(b) Enolate formation

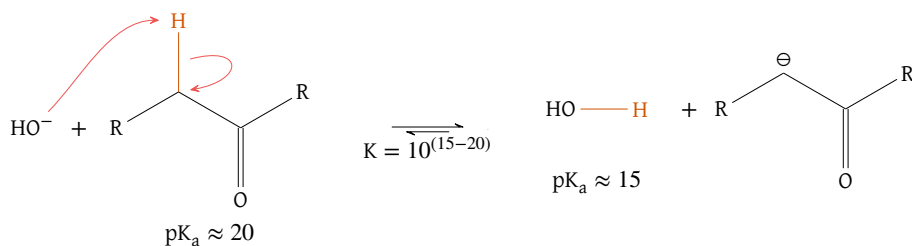


Figure 5: Enolate formation

(c) C-protonation of an enolate

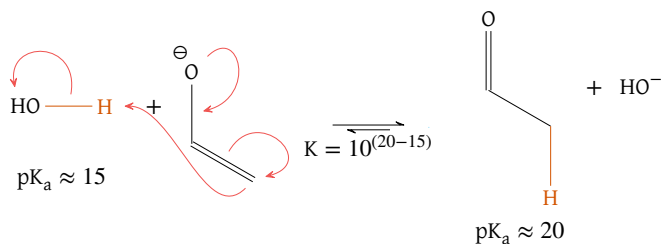


Figure 6: C-protonation of an enolate

(d) Enol formation

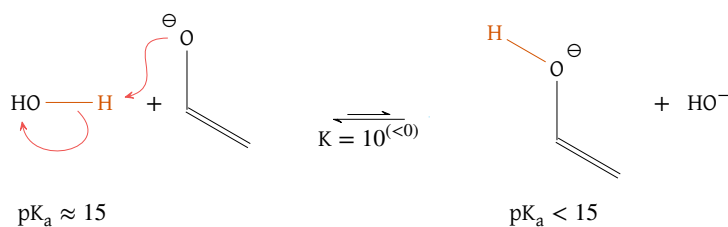
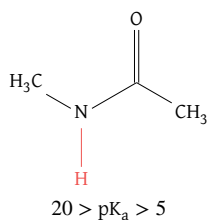


Figure 7: Enol formation

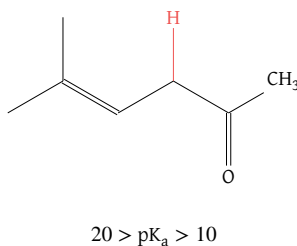
Furthermore, pK_a values of related structures can be predicted from the reference values in Table 1, as demonstrated in the following examples.

(a) Amine like α to carbonyl

Figure 8: Amine like α to carbonyl

The above molecule has a higher pK_a than a carboxylic acid ($pK_a = 5$), since nitrogen is less electronegative than oxygen, but a lower pK_a than a proton α to a carbonyl group ($pK_a = 20$), since nitrogen is more electronegative than carbon.

(b) α to carbonyl and allylic

Figure 9: α to carbonyl group and allylic

Here, The above molecule has a higher pK_a than a proton α to two carbonyl groups ($pK_a = 10$), but a lower pK_a than one α to a single carbonyl group ($pK_a = 20$), since it is additionally allylic.

1.2 Allylic Systems

Core Concept 1.4: Allylic Carbons

Allylic carbons are sp^3 hybridized carbons directly attached to a $C=C$ double bond, and an **allylic hydrogen** is one bonded to such a carbon.

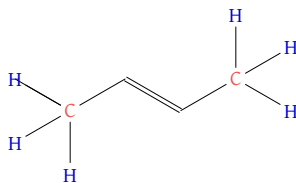


Figure 10: Allylic carbons (red) and allylic hydrogens (blue)

Core Concept 1.5: Allylic System Resonance

An allylic hydrogen exhibits greater acidity than both alkane and alkyne hydrogens, as resonance delocalizes the negative charge across the resulting allylic anion, thereby rendering the anion a comparatively weaker base.

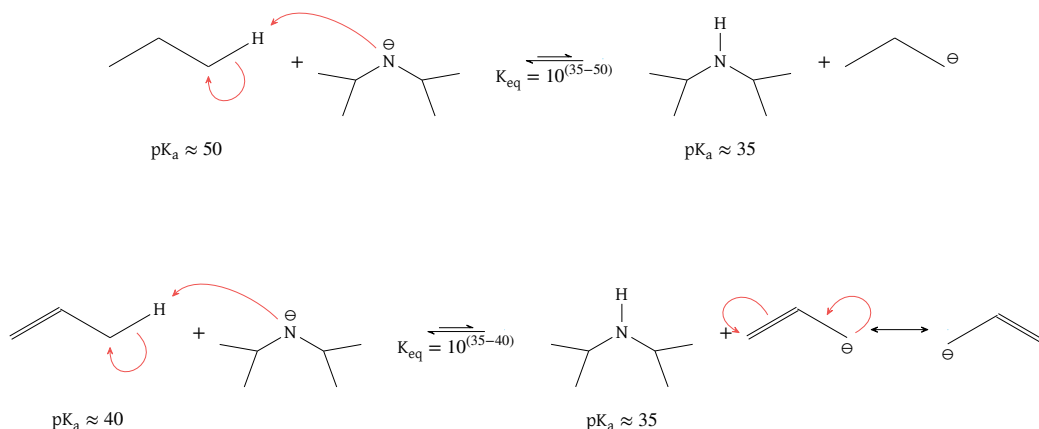


Figure 11: K_{eq} s of alkane (top) vs allylic (bottom) hydrogens; the K_{eq} of the allylic scenario is much more forward favoring because it is able to better stabilize the negative anion state charge

Just as negative charge (filled octets) is stabilized by resonance within allylic structures, radicals and positive charge (unfilled octets) are stabilized through this same resonance mechanism.

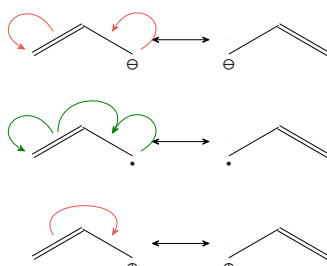


Figure 12: Filled octets (top), radicals (middle), and unfilled octets (bottom)

Critically, **any lone pair capable of participating in resonance must reside in a p orbital.**

Core Concept 1.6: Allylic System Bond Dissociation Energy

As a consequence of resonance-induced stabilization, the bond dissociation energies of distinct hydrogens within an alkene follow a predictable trend.

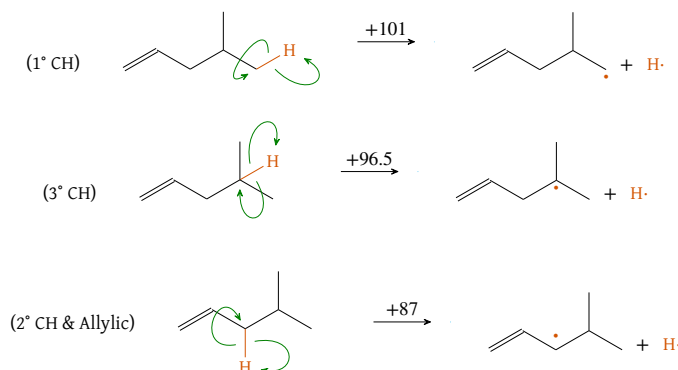


Figure 13: Primary (top), tertiary (middle), and secondary & allylic (bottom) hydrogen bond dissociation energies within the same alkene; secondary allylic hydrogen bond takes least energy to break due to resonance-induced stabilization

Above, the 3° carbon radical is lower in energy than the 1° carbon radical as a result of hyperconjugation between the radical carbon and neighboring σ bonds. The allylic radical, by contrast, receives comparatively less hyperconjugative stabilization as a 2° carbon, yet is stabilized through resonance, causing its corresponding bond dissociation to require the least energy of the three.

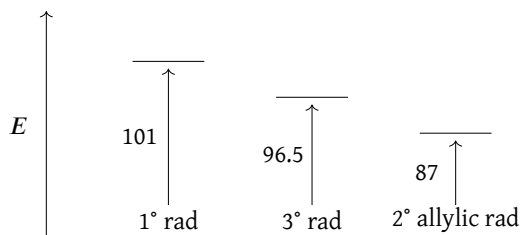


Figure 14: Energy diagram of Figure 13

This differential stabilization gives rise to chemical consequences that play a substantial role in practice, as illustrated by the reaction below.

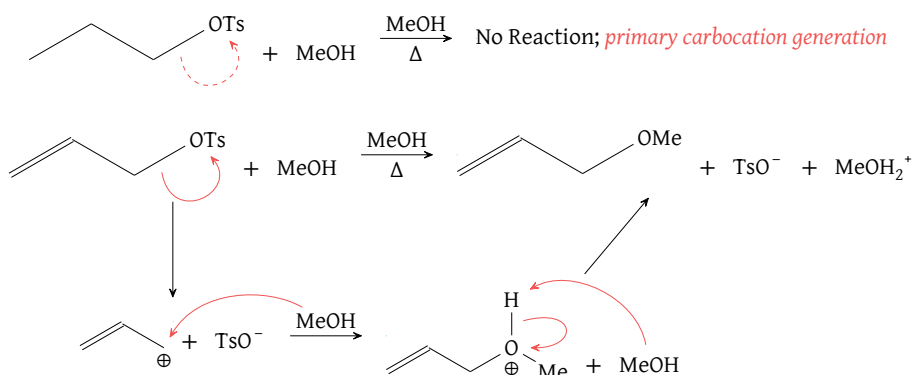


Figure 15: Non-allylic (top) vs allylic (bottom) leaving group primary carbocation formation viability; a primary carbocation can form in the bottom reaction because the cation is additionally allylic; Ts stands for a tosyl group, a common chemical group used in organic synthesis

Core Concept 1.7: Allylic System Molecular Orbital Structure

The molecular orbitals **relevant to allylic structures are the p orbitals**, whose energy levels can be compared in the context of sp - and sp^2 -hybridized carbons, as shown below. Notably, the energy level of sp^2 orbitals lies higher than that of sp orbitals.



Figure 16: Orbital energy level diagrams for sp and sp^2 hybridization

Superimposed on a skeletal diagram backbone, this orbital structure appears as follows.

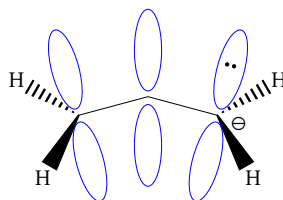


Figure 17: σ bond orbitals overlap along internuclear axis (handshake) whereas π bond orbitals overlap above and below internuclear axes. All carbons are sp^2 hybridized

Considered as a whole, the allylic p-orbital system produces an energy diagram with a distinctive three-level pattern, progressing from bonding to non-bonding to anti-bonding.

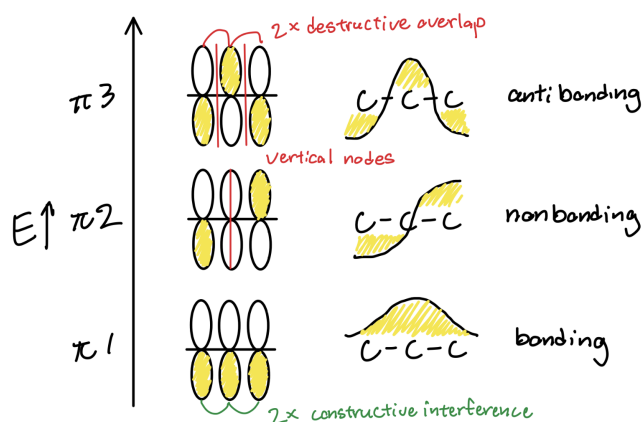


Figure 18: Energy diagram of allylic p-orbital system

Consequently, the HOMO and LUMO assignments differ among the allylic anion, radical, and cation.

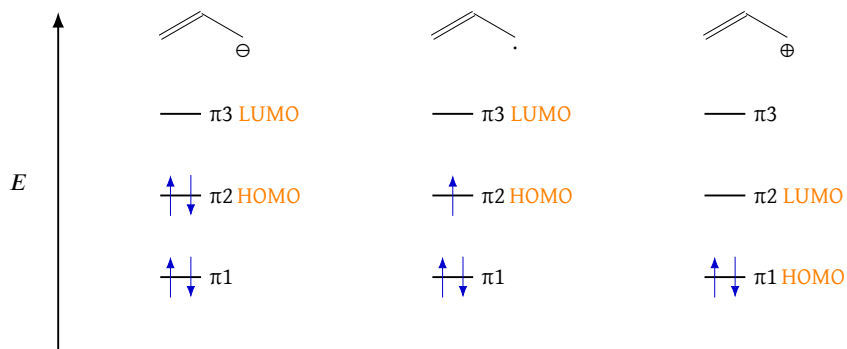


Figure 19: Molecular orbital diagrams for the allyl anion, radical, and cation, showing the filling of π_1 , π_2 , and π_3 with 4, 3, and 2 πe^- s respectively, along with HOMO LUMO assignments

This distinction is particularly relevant in the context of **frontier molecular orbital theory**, which holds that the HOMO of a nucleophile interacts with the LUMO of an electrophile (or vice versa) as explored in the context of Diels–Alder reaction (Core Concept 2.3).

1.3 Conjugated π Systems

Core Concept 1.8: Conjugated π System Stability

A **conjugated π system** refers to a network of adjacent allylic structures that permits resonance delocalization across a greater number of carbons. This effect can be observed by examining the hydrogenation enthalpies of the following alkenes.

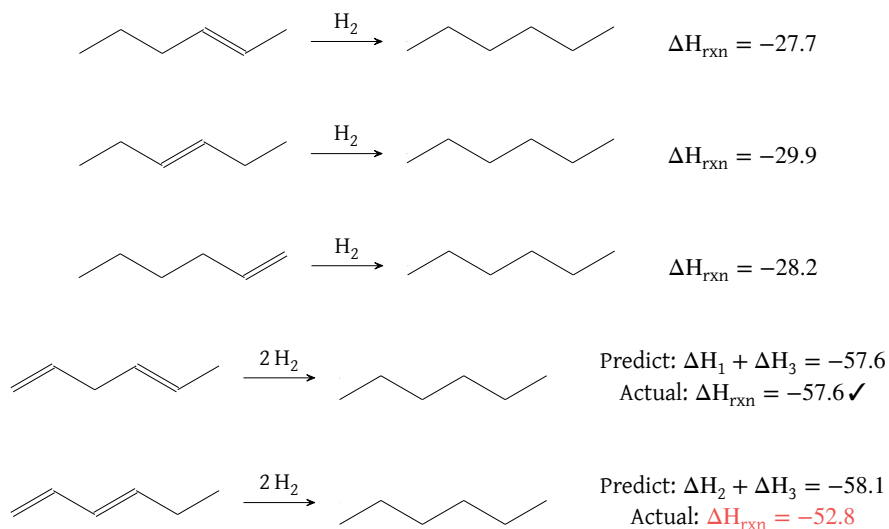


Figure 20: Enthalpies of alkene and alkadiene hydrogenations; the actual enthalpy of a conjugated π system alkadiene is lower than the predicted sum of individual bond hydrogenation enthalpies

Above, compound 5 exhibits greater stabilization than predicted by the simple additive sum of its constituent enthalpies, owing to the enhanced ability of conjugated π systems to stabilize excess or deficient charge.

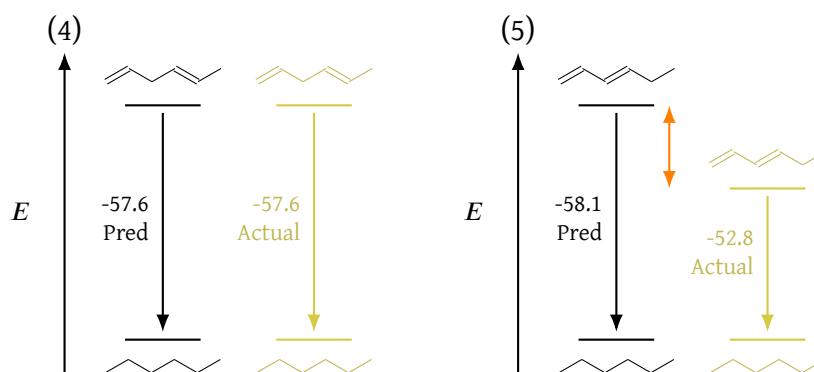


Figure 21: Comparison of predicted versus experimentally measured heats of hydrogenation for a non-conjugated diene (left) and a conjugated diene (right); the conjugated system's smaller actual value reflects a 5 kcal/mol stabilization arising from the four contiguous p -orbitals of the conjugated π system

A closer examination of the orbital energy diagrams for these conjugated π systems, following the same approach applied previously to the three-orbital π system (Figure 18), reveals that the conjugated system's orbital energy levels span an overall larger range while exhibiting smaller stepwise increases in energy between successive levels. Consequently, the HOMO–LUMO gap is comparatively smaller than that of the non-conjugated alkene π system.

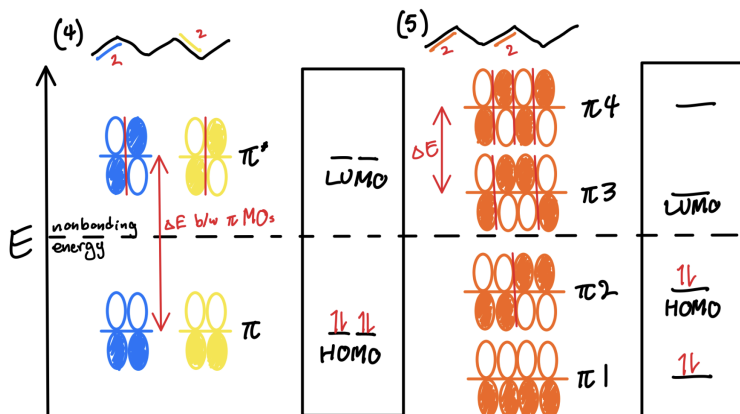


Figure 22: Energy diagrams of non-conjugated (left) and conjugated (right) π systems in alkadienes from Figure 21; source of stability in a conjugated π system are the lowest π MO e^- s

The governing principles for conjugated π system orbital energy diagrams are as follows.

- As conjugation increases (i.e., more p orbitals arranged in succession), the energy gap between π MOs decreases.
- As conjugation increases, the lowest-energy π MO decreases further in energy, while the highest-energy π MO increases further in energy.

It is important to recognize that every high-energy intermediate is stabilized to the greatest extent possible, and that resonance stabilization arising from conjugated π systems supersedes stabilization originating from hyperconjugation.

$$\text{resonance} > 3^\circ > 2^\circ > 1^\circ$$

This principle carries particularly significant consequences for the reactions covered within Alkenes & Alkynes, as many electrophilic additions proceed in either a Markovnikov or anti-Markovnikov manner, a distinction that depends on the relative stability of the intermediate state. Representative examples include alkene hydrohalogenation (Core Concept 1.9), which proceeds in a Markovnikov manner, and hydroboration-oxidation (Core Concept 1.13), which proceeds in an anti-Markovnikov manner.

Applying our understanding of conjugated π system stability, we now consider two illustrative examples.

Core Concept 1.9: Chlorination of Conjugated Cycloalkene

Our objective is to predict all products of the following reaction, including any relevant stereoisomers.

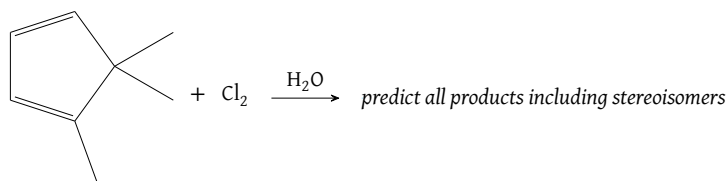
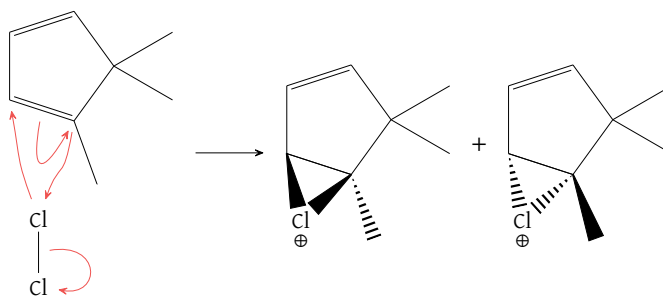
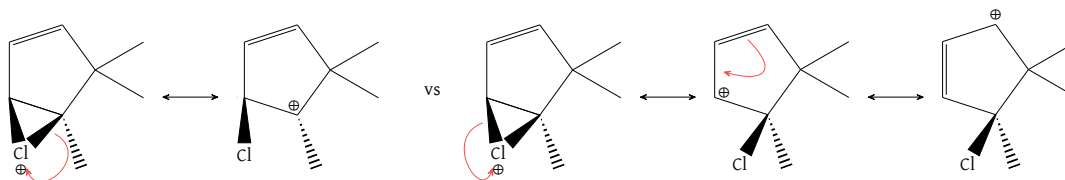


Figure 23: Cycloalkene chlorination reactants

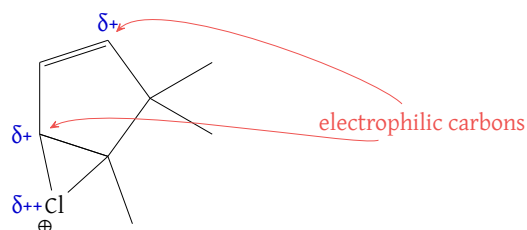
First, we map out the chloronium intermediates.

Figure 24: Top (*left*) and bottom (*back*) attacks leading to corresponding chloronium intermediates

Resonance is used to determine which carbon of the halonium intermediate is preferentially attacked by the weak nucleophile ROH.

Figure 25: Tertiary “ σ bond resonance” (*left*) vs secondary and allylic (*right*)

Because the allylic resonance contributor is more stable, the electrophilic carbons are consequently located at two distinct positions within the system.

Figure 26: Electrophilic carbons on chloronium intermediate; *the top resonance contributor is the Cl*

Consequently, three possible modes of attack exist, giving rise to three distinct stereoisomeric products.

(a) Tertiary electrophilic carbon attack

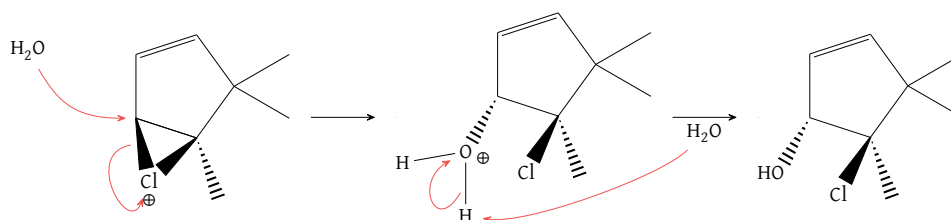


Figure 27: Tertiary electrophilic carbon attack

(b) Secondary electrophilic carbon attack from front

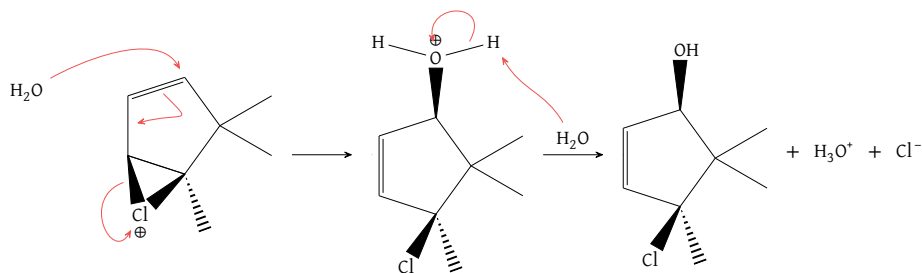


Figure 28: Secondary electrophilic carbon attack from front

(c) Secondary electrophilic carbon attack from back

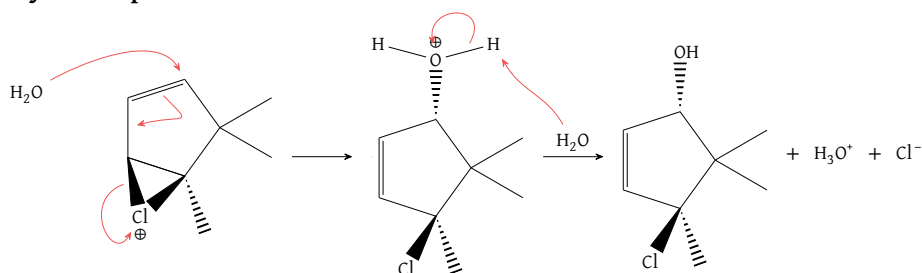
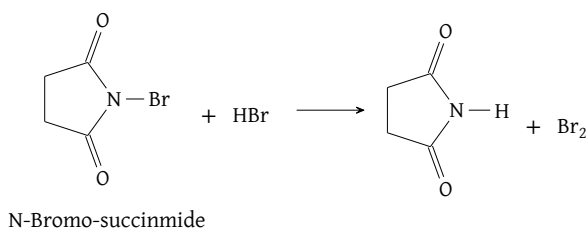


Figure 29: Secondary electrophilic carbon attack from back

Core Concept 1.10: Radical Bromination of Terminal Alkene

The radical bromination of a terminal alkene proceeds in a manner analogous to the reaction shown in Core Concept 1.9, wherein resonance creates the possibility of multiple distinct modes of attack and, consequently, three distinct products.

Br_2 is first generated in situ through **N-Bromosuccinimide (NBS)**, which enables the selective production of **monobrominated products**. It should be noted that direct addition of pure Br_2 to alkenes instead results in dihalogenation, as outlined in Core Concept 1.11.

Figure 30: N-Bromo-succinimide allows to the slow release of Br_2 to avoid alkenes adding Br_2 directly

Following the spontaneous dissociation of Br_2 into radical bromine atoms, the first propagation step occurs, generating a radical on the original alkene compound.

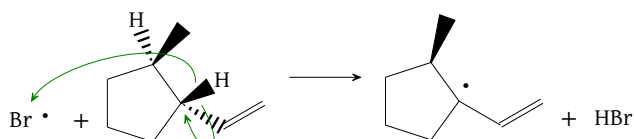


Figure 31: First propagation

This alkene radical and the remaining bromine radical can then undergo a second propagation step in three distinct manners.

(a) **Propagation 2, Tertiary Radical Attack**

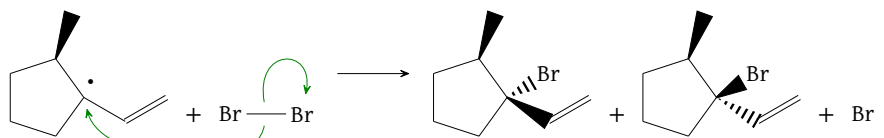


Figure 32: Propagation 2, tertiary radical attack

(b) **Propagation 2, Terminal Radical Attack**

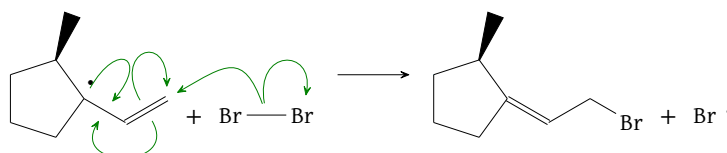


Figure 33: Propagation 2, terminal radical attack

(c) **Propagation 2, Rotated Terminal Radical Attack**

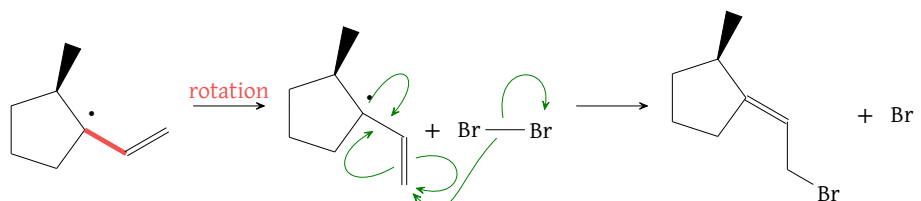


Figure 34: Propagation 2, rotated terminal radical attack

Taken together, the complete reaction proceeds as shown below.

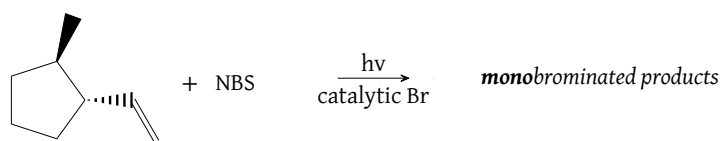


Figure 35: Radical bromination of terminal alkene full reaction

2 Diels–Alder Reactions

Core Concept 2.1: Cycloadditions

Cycloadditions refer to the reaction of two separate π systems to form a new ring, an interaction that involves neither a nucleophile nor an electrophile.

Cycloadditions have already been encountered in two reactions within Alkenes & Alkynes.

(a) **Ozonolysis** (Core Concept 1.18)

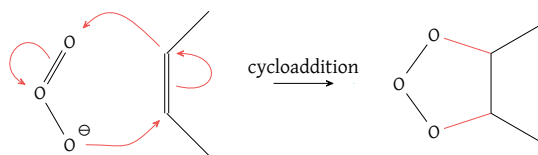
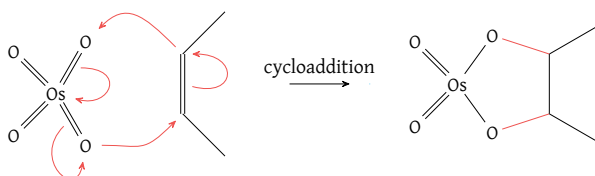
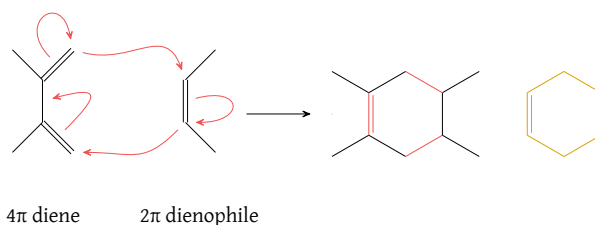


Figure 36: Ozonolysis

(b) OsO_4 Syn-Dihydroxylation (Core Concept 1.17)Figure 37: OsO_4 syn-dihydroxylation

Core Concept 2.2: Diels–Alder Reactions

Diels–Alder reactions are a type of cycloaddition that combine a diene possessing a minimum of 4π e[−]s with a dienophile possessing a minimum of 2π e[−]s, forming a cyclohexene product.

Figure 38: Diels–Alder reaction; combination of a minimum 4π diene with a minimum 2π dienophile

Consequently, diene and dienophile pairings possessing conjugated π systems larger than these stated minimums of 4π and 2π , respectively, can give rise to a substantial diversity of stereoisomeric products.

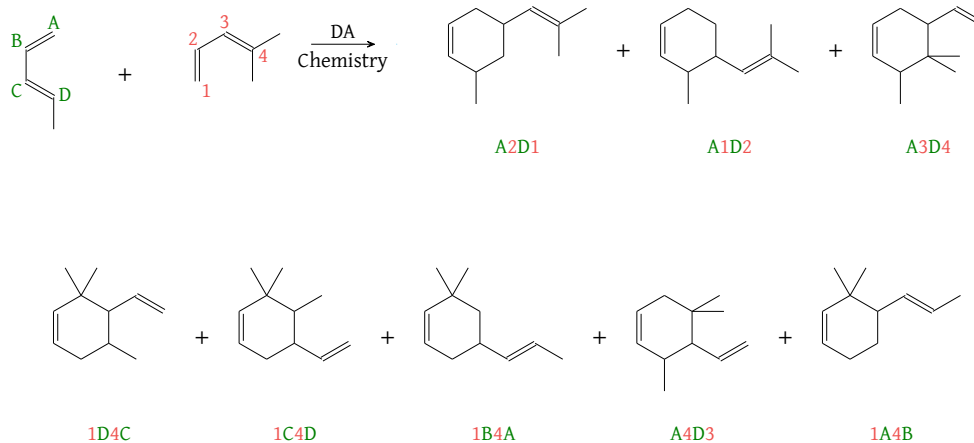


Figure 39: Diels–Alder reaction product diversity

Core Concept 2.3: Orbitals of Diels–Alder Reactions

In Diels–Alder reactions, the π systems of the diene and dienophile approach one another in a vertical orientation, enabling them to generate new σ overlaps.

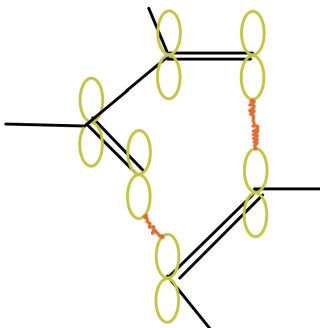


Figure 40: Diene and dienophile vertical interaction; p orbitals overlap to form σ interactions

Consider the energy diagrams of the respective conjugated π systems of the diene and dienophile.

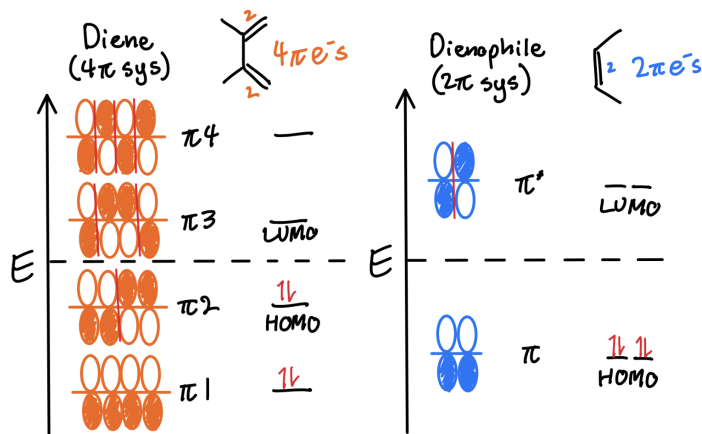


Figure 41: Conjugated π system energy diagrams of a diene and dienophile

A key observation follows from examining which energy levels interact with which: the HOMO of one reagent interacts with the LUMO of the other.

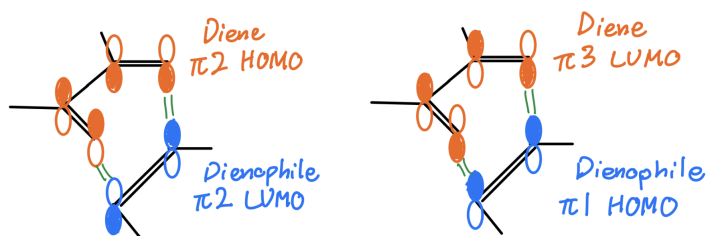


Figure 42: HOMO–LUMO interaction versions between Diels–Alder reagents

Core Concept 2.4: Diels–Alder Stereochemistry

Diels–Alder reactions can further be categorized according to the distinct stereochemical approaches the reagents may adopt.

- (a) **Top** and **bottom** Diels–Alder reactions refer, respectively, to the dienophile approaching the diene from **above** or **below** its plane.



Figure 43: Bottom (left) and top (right) attacks by the dienophile in a Diels–Alder reaction

- (b) **Endo** and **exo** distinctions refer to whether the substituents of the dienophile are oriented **towards** or **away from** the corresponding diene.

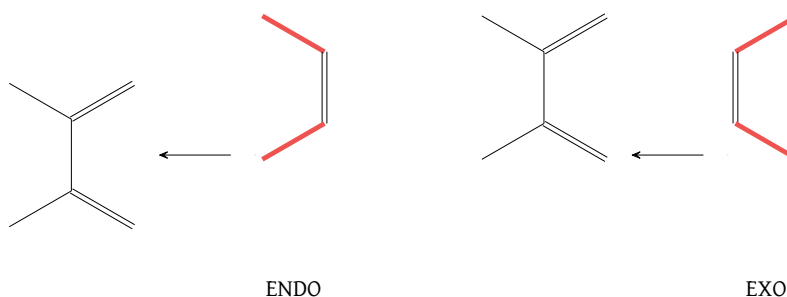


Figure 44: Endo (left) and exo (right) attack by the dienophile in a Diels–Alder reaction

It is important to note that the bonds oriented towards the diene are flipped about the dienophile's double bond, in the direction opposite to the diene's approach.

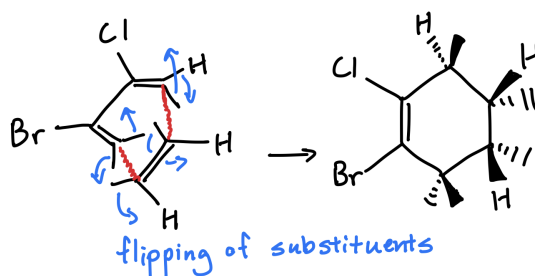


Figure 45: Flipping of dienophile substituents pointing towards the diene during a Diels–Alder reaction

In general, this flipping process can be summarized by the following outline reaction.

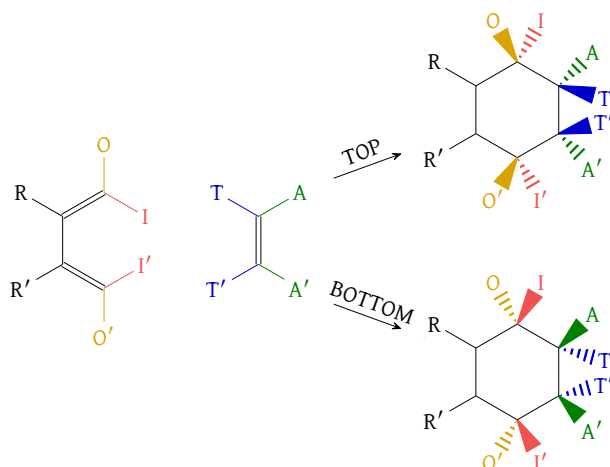


Figure 46: Diastereoselective outcomes resulting from top-face (*top*) versus bottom-face (*bottom*) attack during a cycloaddition reaction

As practice, consider the following Diels–Alder reaction, in which the objective is to predict all non-dimeric Diels–Alder products.

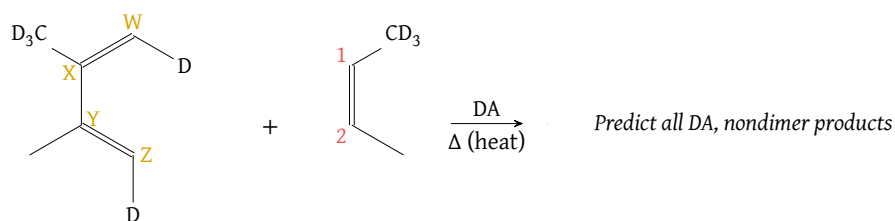


Figure 47: Example Diels–Alder reaction

Two possible constitutional isomers can form, corresponding to the two distinct positionings the dienophile may adopt.

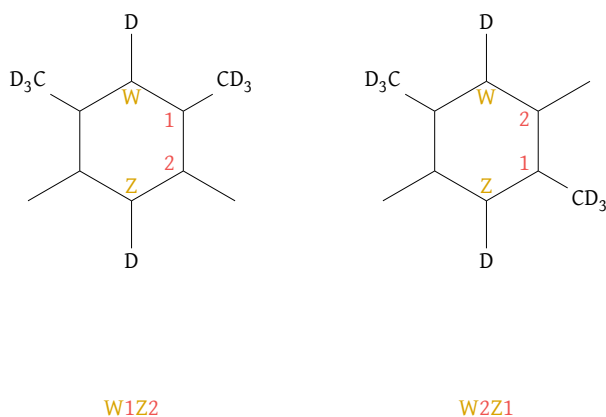


Figure 48: Isomer product possibilities of the reaction shown in Figure 47

It is also important to note that for each of these diene–dienophile arrangements, four distinct stereochemical approaches exist—endo top, endo bottom, exo top, and exo bottom—consistent with the general outline given in Figure 46.

2.1 Rates of Diels–Alder Reactions

Core Concept 2.5: Diene Orientation

Diels–Alder reactions proceed more favorably when the resulting ring exhibits minimal strain. The **s-cis** diene orientation satisfies this condition more effectively than the **s-trans** orientation, as illustrated below. *The “s” in s-cis and s-trans specifically denotes orientation about a single bond, distinguishing this usage from the conventional application of cis/trans nomenclature to alkenes, where the double bond serves as the reference axis.*

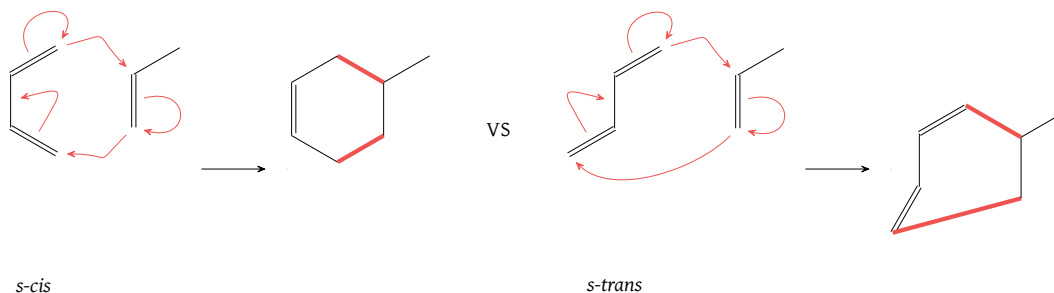


Figure 49: s-cis (left) versus s-trans (right) diene orientation; the s-trans orientation induces substantial ring strain; a ring must contain at least eight atoms to accommodate a trans π bond, and consequently the s-trans reaction depicted above does not proceed

Consider the following dienes as illustrative examples.

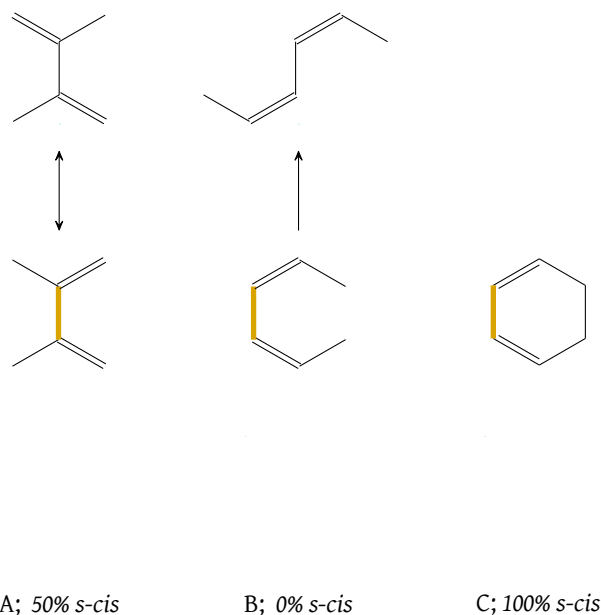


Figure 50: Differences in s-cis orientation prevalence across the three dienes; *Diels-Alder reaction rates favor diene C the most and diene B the least*

The relative Diels–Alder reaction rates follow the order $C > A > B$. Diene C reacts fastest because its cyclic structure restricts rotation about the relevant single bond, thereby preventing any rearrangement into an s-trans conformation. Dienes A and B, by contrast, can both rotate freely about this single bond; however, B favors the s-trans arrangement, since its s-cis conformation experiences comparatively greater steric hindrance between the alkenyl substituents flanking the relevant single bond.

Core Concept 2.6: Secondary Orbital Overlap

Under certain circumstances, orbitals not directly involved in new bond formation between the diene and dienophile can undergo secondary overlap, thereby lowering the overall E_a . Consider the following Diels–Alder interaction between two cyclopentenes.

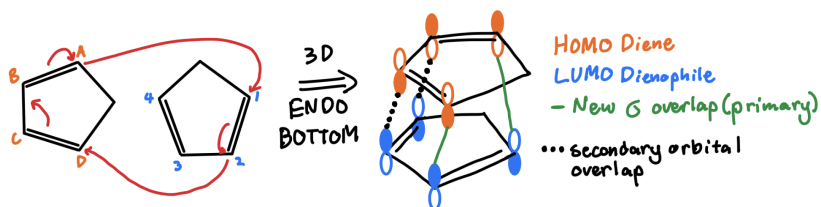


Figure 51: Secondary orbital overlap during an endo Diels–Alder reaction

It is important to note that although favorable mixing of secondary orbital overlaps within the transition state helps stabilize that transition state, steric factors nonetheless disfavor the endo approach. Consequently, ambiguous diene-dienophile pairs typically require experimental investigation to determine which effect predominates in a given case.

Core Concept 2.7: Electronic Effects

Diels–Alder reactions proceed more rapidly when the energies of the interacting orbitals are closer to one another. In other words, the smaller the HOMO–LUMO gap between the reacting molecules, the faster the reaction rate. This condition is typically achieved through increased conjugation.

However, while increased conjugation typically narrows the HOMO–LUMO gap and accelerates the reaction, it also alters the orbital's wave topology (i.e., the arrangement of its nodes). If the orbital possessing the ideal energy gap happens to place a node on an essential bonding atom, that orbital becomes unusable for the reaction, forcing the system to proceed via a higher-energy, slower pathway instead. Symmetry and orbital overlap must therefore always be satisfied before energy gap considerations become relevant.

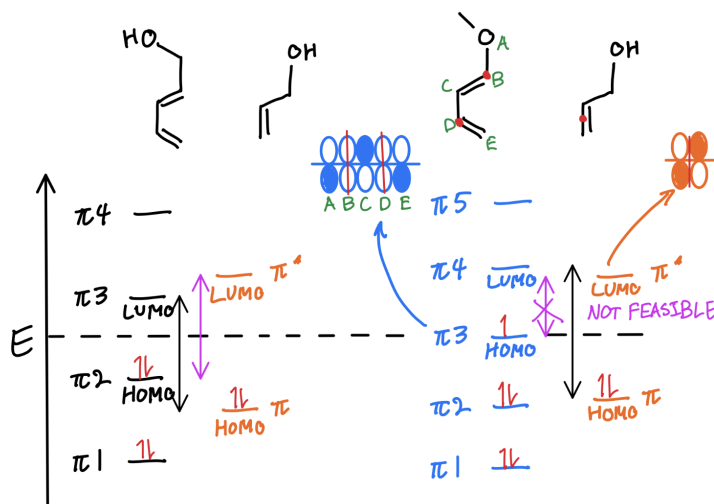


Figure 52: Diels–Alder reaction electronic effects; the reagents on the left actually react faster despite the smaller π_3 – π_4^* gap of the reagents on the right, because the latter possesses a node on the atom required to form a new σ bond

Core Concept 2.8: Groundstate Destabilization

Given that two sets of similar reagents will produce similar transition states leading to the **same** product, the reactant pair whose diene and dienophile begin at a higher ground-state energy level (*due to a factor such as steric hindrance*) will exhibit a faster Diels–Alder reaction rate. Consider the following diene-dienophile pairs, both of which lead to the same product.

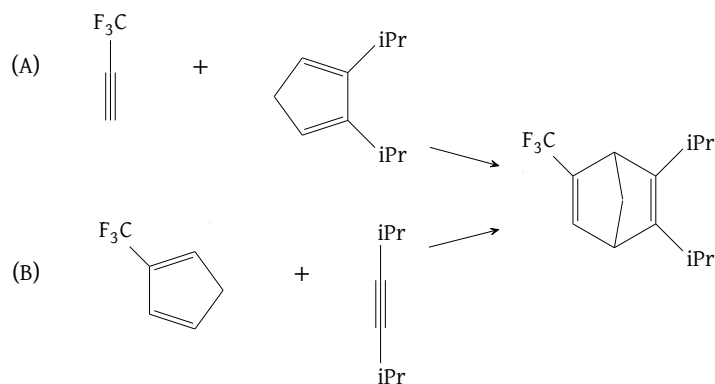


Figure 53: caption

In Figure 53, diene-dienophile pair A possesses a greater degree of steric hindrance between the isopropyl substituents within the diene, placing it at a higher energy level relative to the pair B.

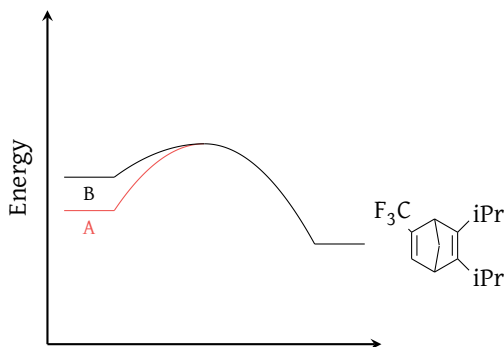


Figure 54: Energy diagrams for the corresponding diene-dienophile pairs from Figure 53; the E_a for compound pair B is substantially smaller than that of pair A, owing to the former's relatively higher starting energy state

As shown in the energy diagram above, the E_a is substantially lower for the more unstable reactant pair A, which consequently proceeds at a much faster rate, since its overall ΔH is lower.

3 Aromaticity & Aromatic Substitution

3.1 Aromatic Compounds

Core Concept 3.1: Aromatic Compounds

The name "aromatic" originates from these compounds' characteristic scent; however, the term now denotes a strictly structural definition, as covered in Core Concept 3.5.

While aromaticity is not limited to the base benzene structure, these notes will primarily refer to benzene-derived compounds such as the following.

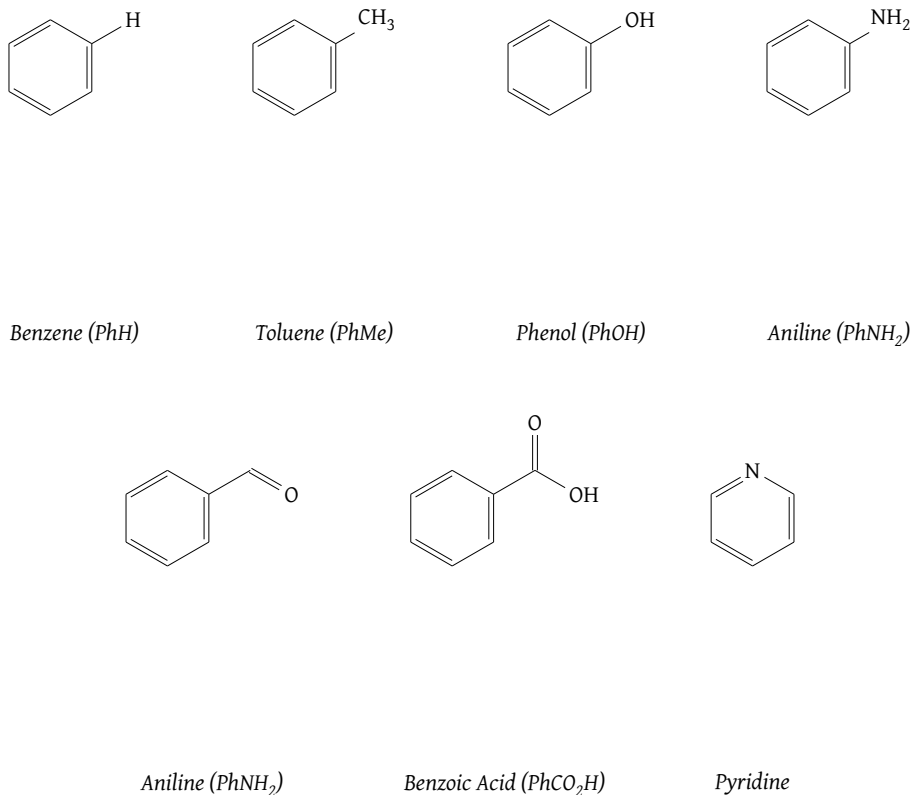


Figure 55: Phenyl ring-based aromatic compound examples; *pyridine* represents an aromatic compound with a non-CH atom as part of its ring

Core Concept 3.2: Aromatic Compound Stereochemistry

Additional substituent positions on the root aromatic compounds shown above can be specified using the ortho, meta, and para keywords.

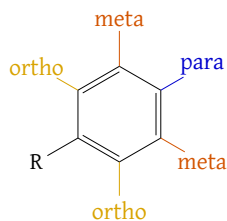
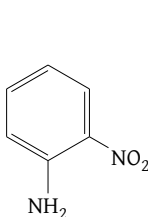
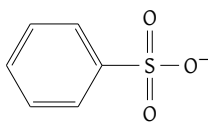


Figure 56: Phenyl ring substituent-positioning keywords

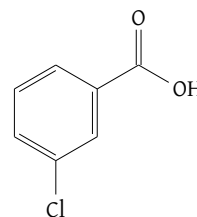
The compound naming scheme follows the structure *relation-substituent-root*. Consider the following examples.



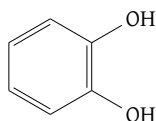
ortho-nitro-aniline



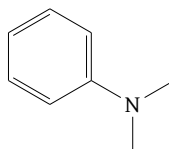
para-sulfonate-toluene



meta-chloro-benzoic acid



ortho-hydroxyl-phenol



para-dimethylamino-pyridine

Figure 57: Phenyl ring-based aromatic compound nomenclature

Core Concept 3.3: Reactivity of Aromatic Compounds

Aromatic compounds are considerably less reactive than would otherwise be predicted. For instance, while a simple cyclohexene reacts readily with mCPBA to form an epoxide, no reaction occurs when benzene is mixed with three equivalents of mCPBA.

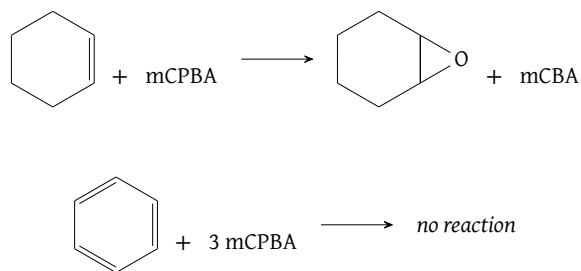


Figure 58: Reactions of cyclohexene and benzene with mCPBA

Similarly, a substantial portion of the predicted energy release during the hydrogenation of benzene is offset by aromatic stabilization.

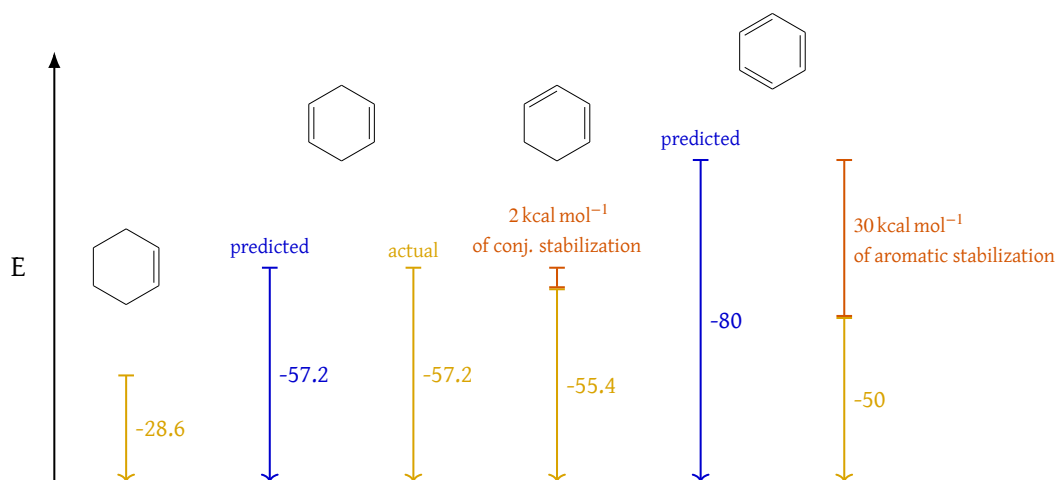


Figure 59: Energy diagram of predicted versus actual heats of hydrogenation for cyclohexene derivatives; *aromaticity confers a substantial increase in stability relative to the summed values of the corresponding isolated double-bond hydrogenations, exceeding even the stabilization observed in nonaromatic but conjugated cyclohexene systems*

To explain this phenomenon, consider a top-down view of the benzene π molecular orbital diagram.

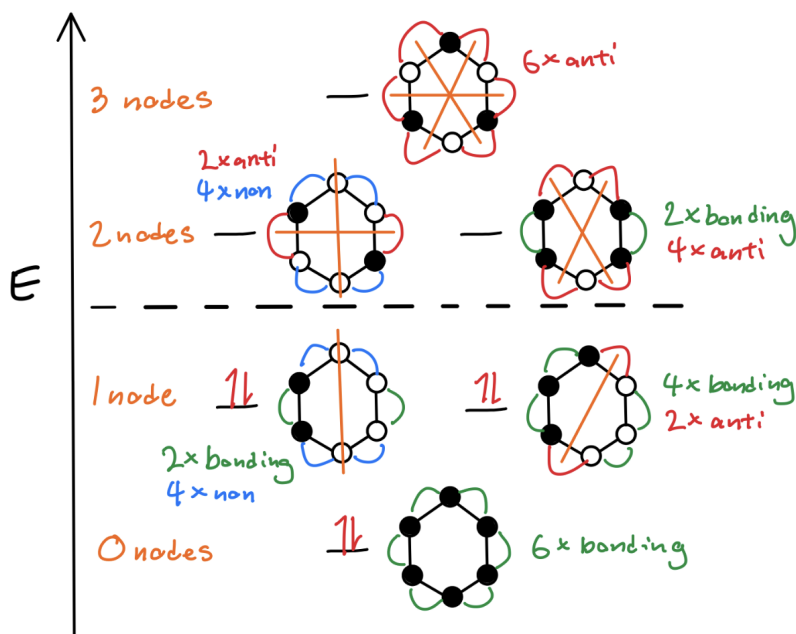


Figure 60: Energy diagram of the benzene π molecular orbitals (top-down view)

As shown in Figure 60, the π electrons occupy exclusively bonding molecular orbitals, rendering the system exceptionally stable.

Core Concept 3.4: Aromatic Compound Molecular Diagrams

The **Frost Diagram** is a method for rapidly predicting π molecular orbital energy levels of a cyclic conjugated system, following the steps below.

- (1) The compound must possess a cyclic array of p orbitals
- (2) Draw the ring structure with one vertex pointed downwards
- (3) Draw a circle around the compound
- (4) Intersections represent energy levels
- (5) The middle of the circle represents the nonbonding energy level
- (6) Add e^- s

Below are the corresponding orbital diagrams for a cyclopentenyl anion and a cycloheptenyl cation; the benzene diagram has already been shown above.

(a) Cyclopentene Anion

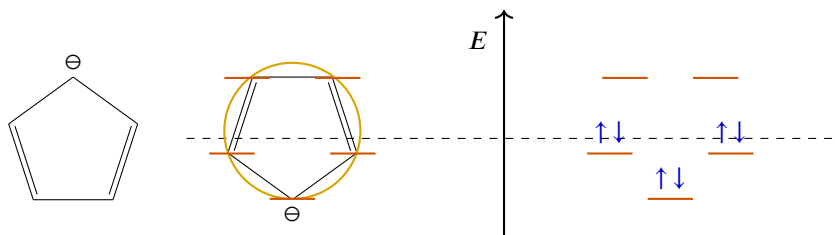


Figure 61: Frost diagram for the cyclopentenyl anion, showing three filled bonding/nonbonding orbitals and two unoccupied, degenerate antibonding orbitals

(b) Cycloheptene Cation

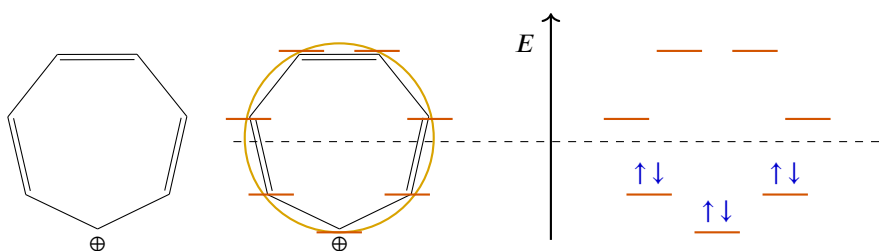


Figure 62: Frost diagram for the cycloheptenyl cation, showing three filled bonding/nonbonding orbitals and four unoccupied, degenerate antibonding orbitals

Core Concept 3.5: Aromaticity Rules

In order for a compound to be considered aromatic, it must satisfy a strict set of rules.

- (1) The compound must be a cyclic array of p orbitals
- (2) The e^- count of the cyclic array must fulfill the equation of $4n + 2$, where n is an integer (0, 1, 2, 3, 4, ...).
*An e^- count that instead fulfills the equation $4n$, where n is an integer, results in a compound that is far more reactive than predicted and is termed **anti-aromatic**.*

If the above criteria are not met, the compound is considered **nonaromatic**, as in the cyclohexene shown below.

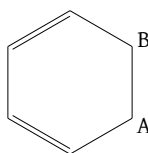


Figure 63: Nonaromatic cyclohexene; carbons A and B are sp^3 , which prevents an uninterrupted cyclic array of p orbitals

As practice, consider the following two examples.

(a) Pentene Anion

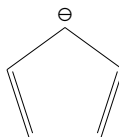


Figure 64: Aromatic pentene anion

This structure exhibits a cyclic array of p orbitals with an electron count of 6, satisfying the $4n + 2$ expression. This compound is therefore **considered aromatic**.

(b) **Pentene Cation**

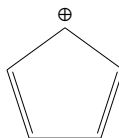


Figure 65: Anti-aromatic pentene cation

This compound likewise exhibits a cyclic array of p orbitals, but its electron count of 4 satisfies $4n$ rather than $4n + 2$. This compound is therefore considered **anti-aromatic**.

Notice that the corresponding molecular orbital diagram results in two unpaired electrons, accounting for its heightened reactivity.

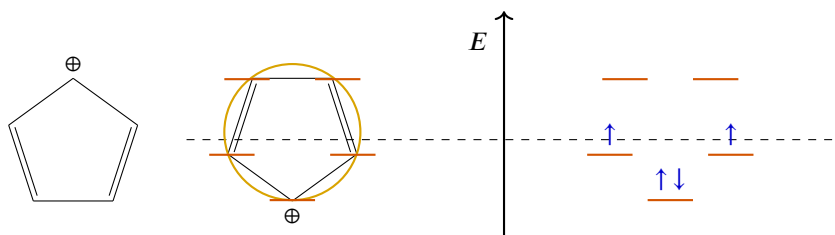


Figure 66: Frost diagram for the cyclopentenyl cation

Core Concept 3.6: Aromatic Contributors

An **aromatic contributor** refers to a resonance structure of a molecule that possesses a fully conjugated, cyclic, planar π system satisfying the $4n + 2$ π -electron count, even when the dominant or lowest-energy Lewis structure does not immediately appear aromatic. Consider the following molecule as an example.

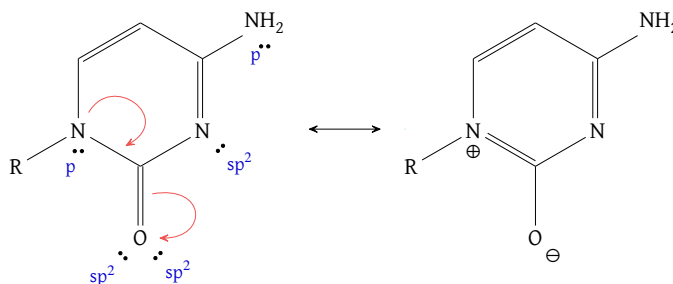


Figure 67: Aromatic contributor; a simple rule of thumb for identifying a resonance contributor is to attempt to achieve alternating π bonds while ensuring that a circle drawn around the ring never crosses a π bond

Notice that although the base resonance structure is nonaromatic by definition, since its conjugated π orbital system is not cyclic, an alternative structure accessible through resonance is aromatic, thereby qualifying the molecule as possessing an aromatic contributor.

Core Concept 3.7: Aromaticity Exceptions & Basicity

While most aromatic systems conform to the criteria outlined in Core Concept 3.5, certain ring structures constitute notable exceptions.

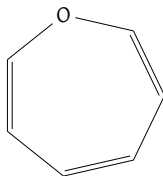


Figure 68: Oxepin; although its p orbitals appear to form a cyclic array satisfying the $2n$ electron count associated with anti-aromaticity, experimental data show oxepin to be non-aromatic

An initial assessment of oxepin's (Figure 68) π system suggests a total of eight p -orbital electrons, six contributed by the three double bonds and two by the oxygen's lone pair, which would nominally classify the molecule as anti-aromatic. However, because anti-aromaticity imposes a severe thermodynamic penalty, the molecule instead avoids this unstable configuration by disrupting the continuous cyclic array required for it. It accomplishes this by **localizing the oxygen's lone pair and excluding it from resonance**. By breaking the conjugated loop of p orbitals in this manner, oxepin renders itself **non-aromatic rather than anti-aromatic**.

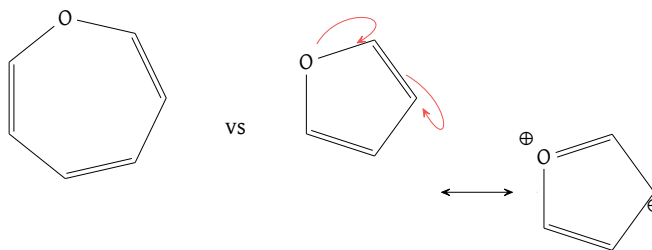


Figure 69: Comparison of resonance behavior in oxepin (left) versus furan (right)

This structural anomaly carries direct consequences for basicity, as illustrated by the comparison between oxepin and furan (Figure 69).

- In **oxepin**, the oxygen's lone pair is strictly localized in order to prevent the continuous cyclic array that would otherwise produce an unstable anti-aromatic state. Because these electrons remain confined to the oxygen and do not participate in resonance, they stay fully available to accept a proton.
- In contrast, the oxygen's lone pair in **furan** delocalizes actively into the ring's π system. This resonance withdraws electron density from the oxygen, imparting a partial positive charge and substantially diminishing its capacity to act as an electron donor.

Consequently, the forced localization of the lone pair in oxepin renders its oxygen atom significantly more basic than the resonance-delocalized oxygen of the six-membered furan ring.

3.2 Electrophilic Aromatic Substitution

Core Concept 3.8: Electrophilic Aromatic Substitution

Electrophilic aromatic substitution (EAS) describes a class of reactions in which an aromatic ring functions as the nucleophilic species. The general mechanism underlying this reaction type is outlined below.

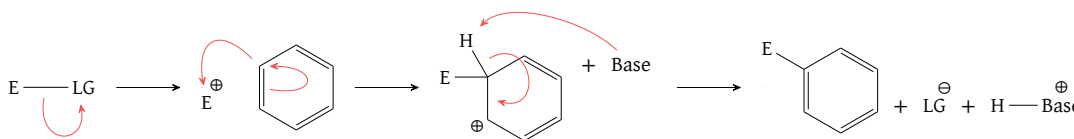


Figure 70: General mechanistic outline of electrophilic aromatic substitution reactions

3.2.1 Reaction Types

Core Concept 3.9: Halogenation

EAS halogenation results in the attachment of a single halogen atom to an aromatic ring, with the corresponding hydrohalic acid generated as a byproduct. This mechanism proceeds via activation by a **Lewis acid** (defined as any chemical species capable of accepting an electron pair), as illustrated below.

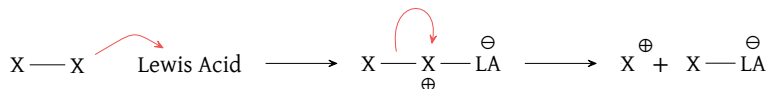


Figure 71: Generation of a halogen cation electrophile via Lewis acid activation; *Lewis acids function as electron-pair acceptors and include species such as FeX_3 , BX_3 , and AlX_3*

The chlorination and bromination of benzene, presented below, illustrate this general mechanism.

(a) Chlorination of Benzene

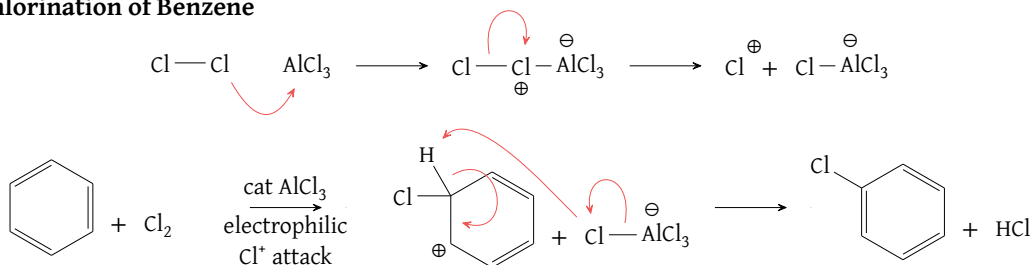


Figure 72: Chlorination of benzene

(b) Bromination of Benzene

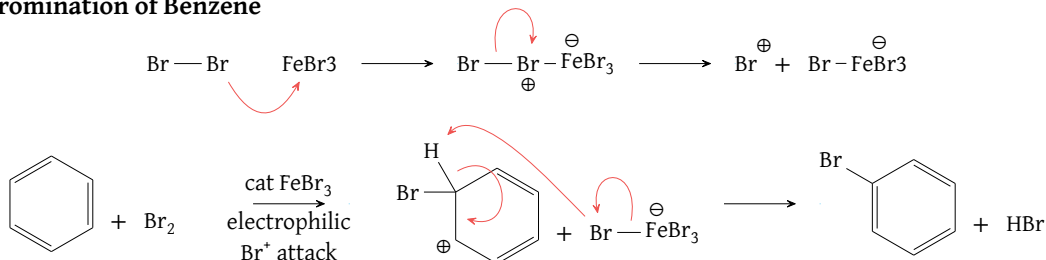
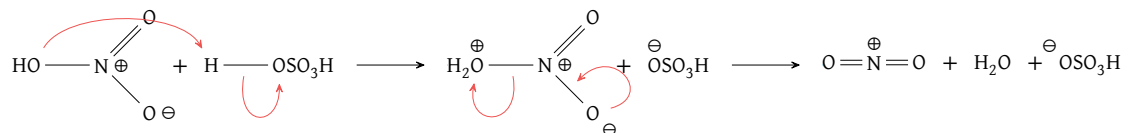


Figure 73: Bromination of benzene

Core Concept 3.10: Nitration

Nitration involves the attachment of a nitro group (NO_2) to an aromatic ring and serves primarily as a synthetic intermediate step in practice. Consider first the generation of the NO_2^+ electrophile, formed upon exposure of nitric acid to sulfuric acid.

Figure 74: Generation of the NO_2^+ electrophile

With this electrophile established, the nitration of benzene proceeds according to the general mechanism outlined in Core Concept 3.8.

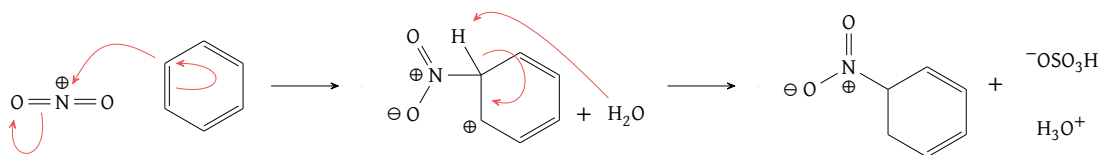
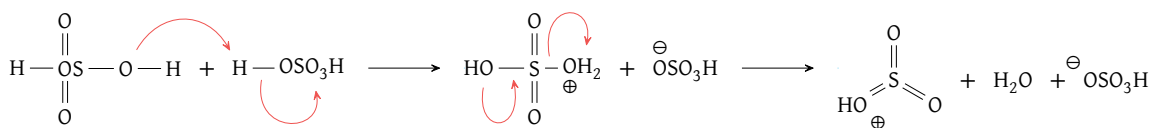


Figure 75: Nitration of benzene

Core Concept 3.11: Sulfonation

Sulfonation entails the attachment of a sulfonic acid group (SO_3H) to an aromatic ring. The initial step involves a reaction between two molecules of H_2SO_4 , arising from the use of **fuming H_2SO_4** as one of the reagents.

Figure 76: Formation of fuming H_2SO_4

The reaction then proceeds according to the same general EAS mechanism described in Core Concept 3.8.

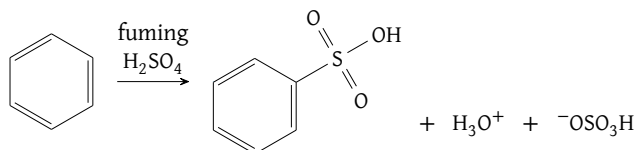


Figure 77: Sulfonation of benzene

Core Concept 3.12: Alkylation

Alkylation refers to the addition of an alkyl group to an aromatic structure. Before an alkane can act as an electrophile, it must first undergo conversion to a carbocation intermediate, as shown in the reactions below.

(a) Generation of an Electrophile via a Lewis Acid

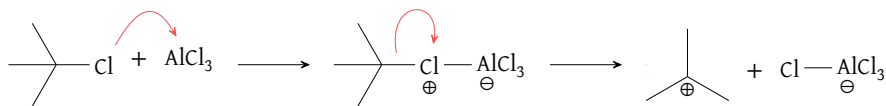


Figure 78: Mechanism for the formation of a tert-butyl carbocation from tert-butyl chloride using aluminum chloride (AlCl_3) as a Lewis acid catalyst

This transformation parallels the initial step of the chlorination pathway presented in Core Concept 3.9.

(b) Lewis Acid-Catalyzed Epoxide Ring Opening

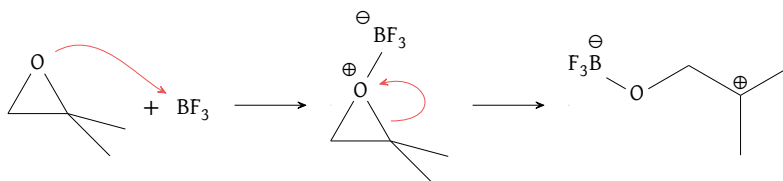


Figure 79: Boron trifluoride (BF_3) coordination to an epoxide, promoting regioselective ring cleavage to generate the more thermodynamically stable tertiary carbocation

(c) Acid-Catalyzed Carbocation Formation from an Alcohol

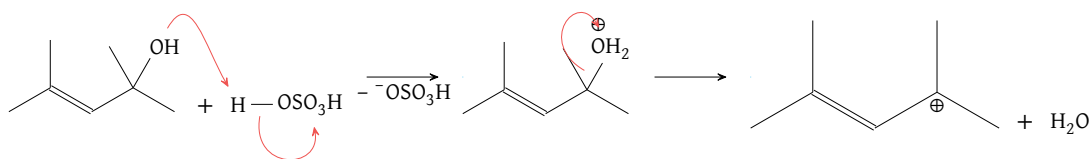


Figure 80: Protonation of a tertiary alcohol by sulfuric acid, followed by the expulsion of water to form a tertiary carbocation intermediate

(d) Protonation of an Aromatic Ring

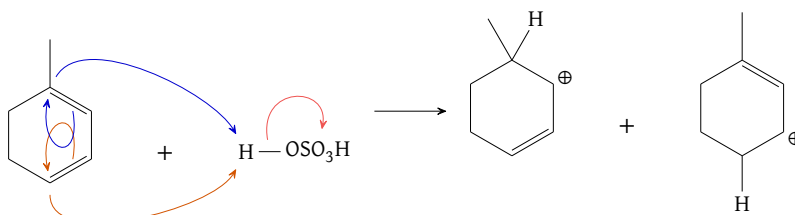


Figure 81: Electrophilic attack on toluene by H^+ ; the first hydrogenation step follows either the blue or orange e^- movement

In certain cases, carbocation rearrangement occurs when the initially formed carbocation is unstable.

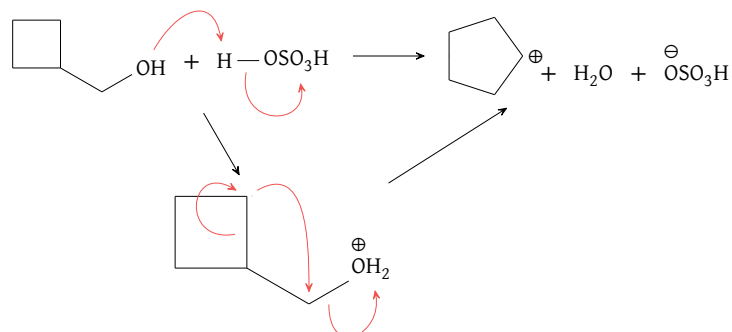


Figure 82: Acid-catalyzed dehydration and subsequent ring expansion to generate a more thermodynamically stable cyclopentyl carbocation

These rearranged carbocations subsequently undergo EAS with aromatic structures.

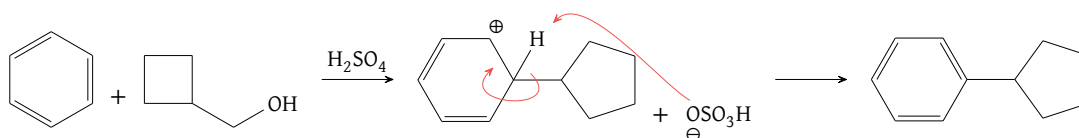


Figure 83: Electrophilic aromatic substitution of benzene by a rearranged cyclopentyl carbocation

In the case above, illustrating rearrangement from a cyclobutyl to a cyclopentyl carbocation, such rearrangement can be avoided through a stepwise pathway involving **acylation** (Core Concept 3.13) followed by reduction with Zn(Hg) and HCl.

(a) **Acylation**

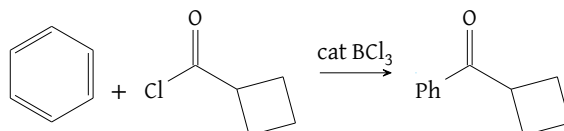


Figure 84: Acylation

(b) **Reduction**

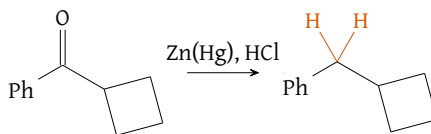
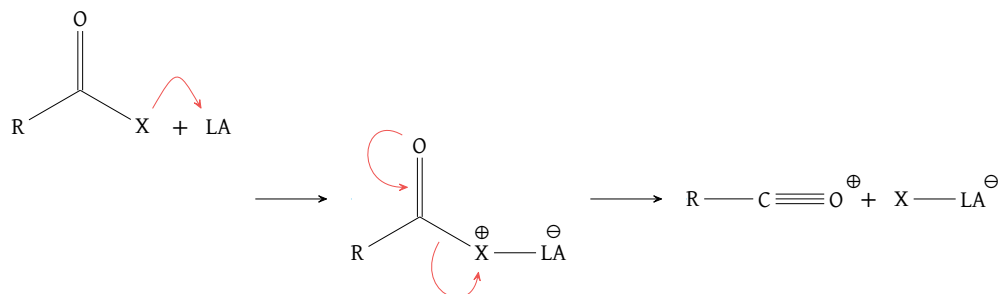


Figure 85: Reduction of ketone with Zn(Hg) and HCl

Core Concept 3.13: Acylation

Acylation refers to the addition of an acyl halide to an aromatic structure. The acyl halide must first be rendered electrophilic through activation by a Lewis acid, as shown in the general reaction outline below.



Acyl halide

Figure 86: Conversion of an acyl halide to an electrophile

The resulting RCO^+ species can then undergo EAS with an aromatic ring, as demonstrated below using representative compounds.

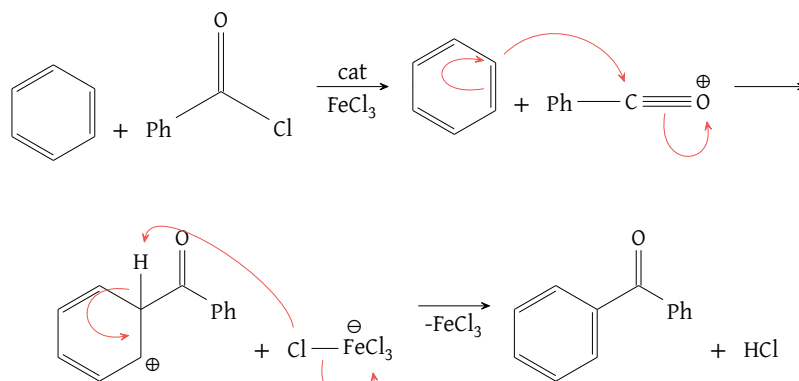


Figure 87: Acylation of benzene

3.2.2 Directing Groups

Core Concept 3.14: Directing Groups

So far, we have covered the EAS reaction types in Subsubsection 3.2.1 using benzene as our sole relevant aromatic structure. However, in scenarios where the aromatic ring is asymmetrical, chemists require a method for determining which carbon the EAS reaction will occur on. We now introduce **directing groups**, functional groups already attached to a benzene ring that dictate where (*ortho*, *meta*, *para*) an incoming electrophile will land.

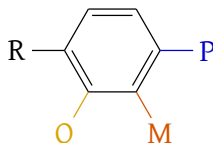


Figure 88: Directing groups ortho (yellow), meta (orange), and para (blue)

The table below summarizes directing group behavior; consult each Core Concept entry for a more detailed treatment of each type.

Directing Group Type	Examples	Description
(1) Inductive O/P Director	$R = \text{alkyl [Me, Et, } i\text{Pr, ...]}$	Alkyl groups donate electron density inductively, directing substitution to the ortho and para positions.
(2) Inductive M Director	$R = \text{carbon w/ EN atoms [CCl}_3, \text{CF}_3, \text{...}]}$	Carbons bearing electronegative atoms withdraw electron density inductively, directing substitution to the meta position.
(3) Resonance O/P Director	i) R has lone pair adjacent to ring: [OMe, SR, NMe ₂] ii) $R = \text{carbon w/ } \pi \text{ system: [alkene, alkyne, benzene]}$	Substituents that donate electron density via resonance into the ring direct substitution to the ortho and para positions.
(4) Resonance M Director	[C=O, C≡N, NO ₂]	Substituents that withdraw electron density via resonance from the ring direct substitution to the meta position.

Table 2: Classification of electrophilic aromatic substitution directing groups

Core Concept 3.15: Inductive O/P Directors

Inductive O/P directors are substituents that direct any EAS to the ortho and para positions within an aromatic ring.

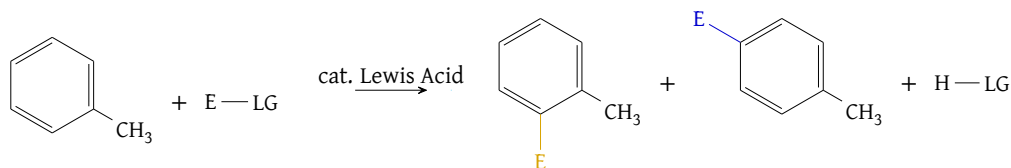
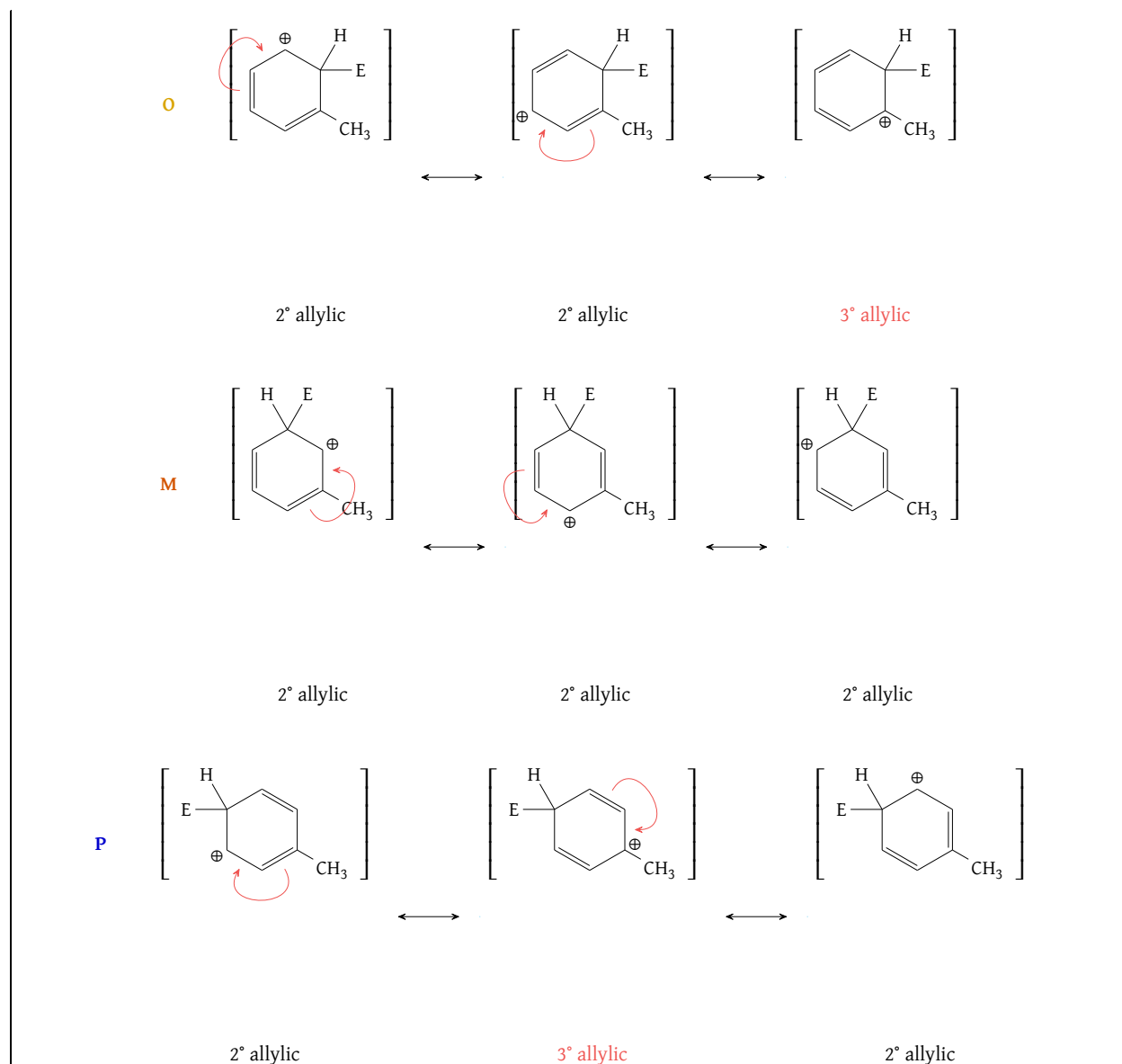


Figure 89: Inductive O/P directors

These substituents are typically alkyl groups; their preference for O/P over M positioning arises from the relative stability of the resonance structures generated in each case.

Figure 90: Ortho, meta, para positioning resonance structures of H_3CPh

As shown above in Figure 90, although most resonance structures place the transient carbocation charge at a secondary allylic position, O/P substitution permits a tertiary allylic carbocation in one of three resonance structures, whereas M substitution does not. Consequently, the relative energies of the resulting carbocation intermediates follow the ordering shown in the energy diagram below.

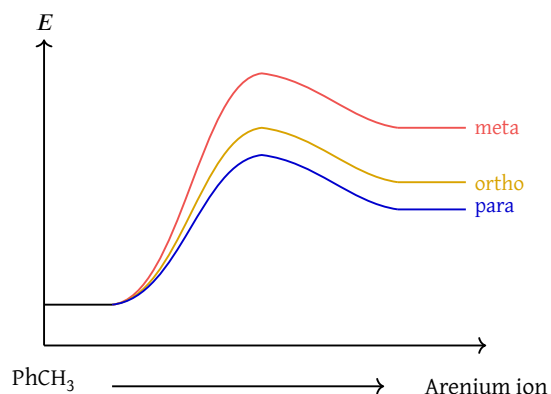


Figure 91: Energy diagram comparing the relative energies of the meta, ortho, and para transition states leading from the reaction $\text{H}_3\text{CPh} + \text{E}^+ \longrightarrow \text{H}_3\text{CPhE}^+$

Core Concept 3.16: Inductive M Directors

Inductive M directors are substituents that direct EAS to the meta position within an aromatic ring. These substituents consist of carbons bonded to highly electronegative atoms, rendering the carbon strongly partially positive.

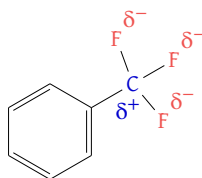


Figure 92: Inductive M directors

The M position is favored because the O/P 3° allylic carbocation resonance structures experience destabilizing repulsion between the carbocation and the partially positive substituent carbon. The M position, by contrast, permits a resonance structure in which the positive substituent carbon and the carbocation charge reside at maximal distance on opposite sides of the aromatic ring.

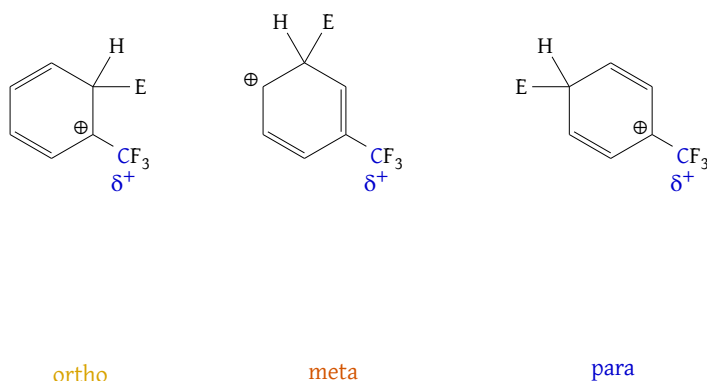


Figure 93: Best resonance contributors of each position in F_3CPh ; the O/P 3° allylic contributors experience additional Coulombic **repulsion** due to the partially positive adjacent carbon of the CF_3 substituent

Consequently, the resulting energy levels place O/P positioning above that of M positioning.

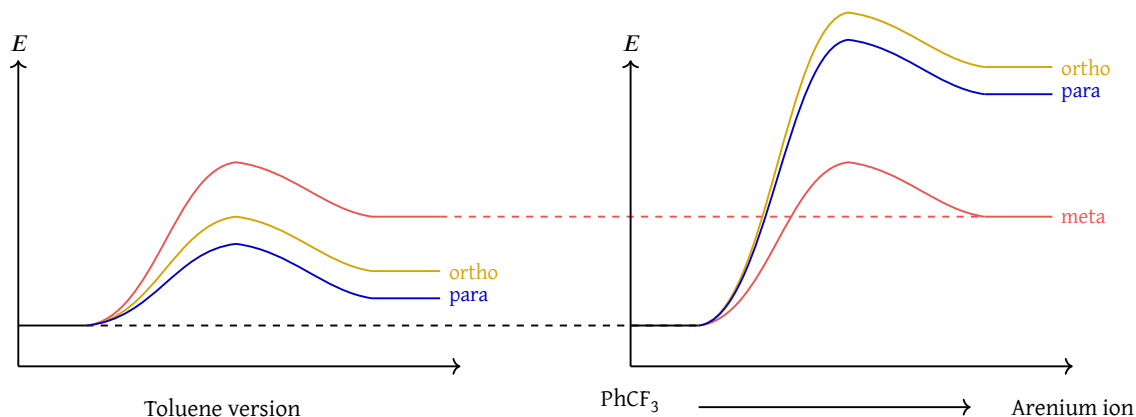


Figure 94: Energy diagram comparing the transition-state energies for meta, ortho, and para substitution in the reactions $\text{H}_3\text{CPh} + \text{E}^+ \longrightarrow \text{H}_3\text{CPhE}^+$ and $\text{F}_3\text{CPh} + \text{E}^+ \longrightarrow \text{F}_3\text{CPhE}^+$; inductive directors raise the O/P energies comparatively more, making meta the relatively favored position

Core Concept 3.17: Resonance O/P Directors

Resonance O/P directors are substituents that direct EAS to the O and P positions within an aromatic ring by donating e^- density. Phenol, for example, possesses resonance structures that localize the resulting carbanion charge exclusively at the O and P positions.

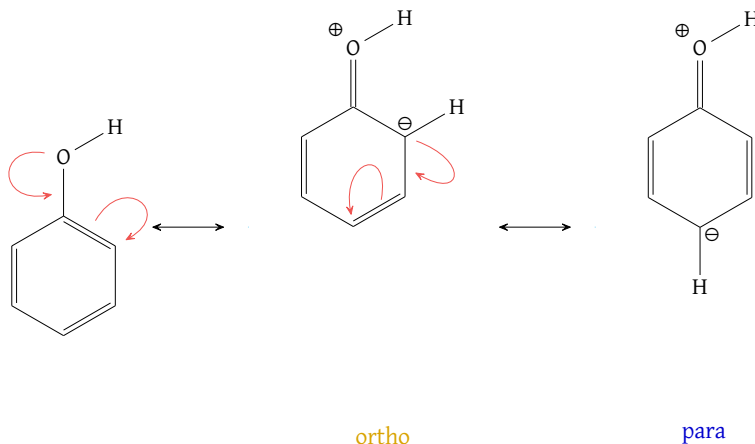
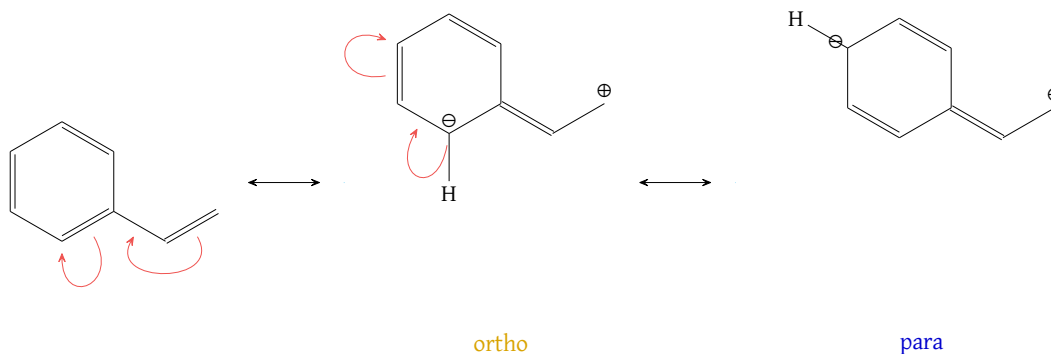


Figure 95: Resonance O/P director; this mechanism requires a substituent capable of donating an electron pair

These substituents must specifically possess a lone pair adjacent to the aromatic ring. Another substituent type capable of e^- donation is a carbon bearing a conjugated π system, as illustrated by the following alkene.

Figure 96: Conjugated π system resonance O/P directing

Note that formation of the primary carbocation is favorable here because it is allylic.

As a side note on how inductive and resonance directors differ in the mechanism by which they induce positional favorability: resonance directors alter the relative nucleophilicity of the ring positions, whereas inductive directors influence favorability through transition-state stability.

Core Concept 3.18: Resonance M Directors

Resonance M directors are substituents that direct EAS to the M position within an aromatic ring by withdrawing e^- density. Carbonyl substituents, for example, withdraw e^- density such that the resulting resonance structures place a positive charge at the O/P positions, rendering these positions less nucleophilic and the M position relatively more nucleophilic.

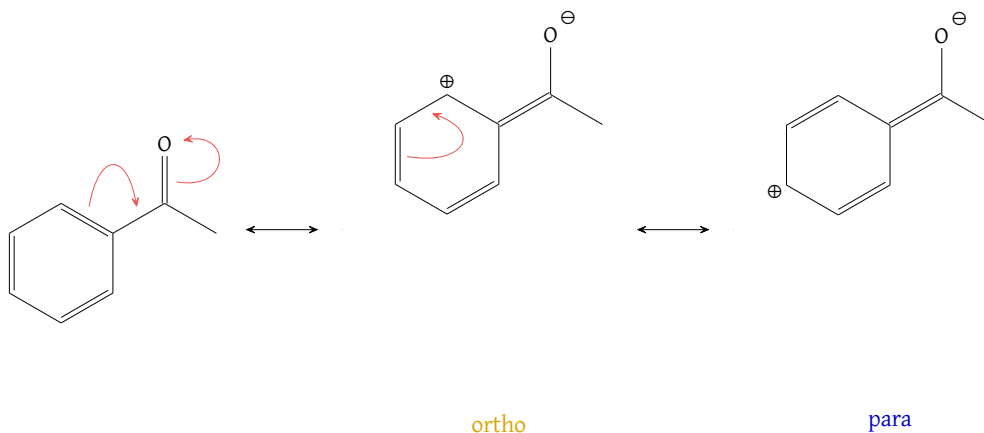


Figure 97: Resonance M director

As a result, when an electrophile approaches the transient aromatic ring, nucleophilic attack occurs most favorably from the M position rather than the O/P positions.

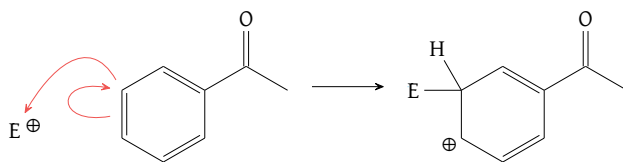


Figure 98: Electrophilic aromatic substitution in the presence of resonance M directors

Core Concept 3.19: Halogens as Directing Groups

Halogens can also function as directing groups; empirically, they act as O/P directors. To understand the basis for this behavior, consider monohalobenzene in the context of both the inductive effect and resonance.

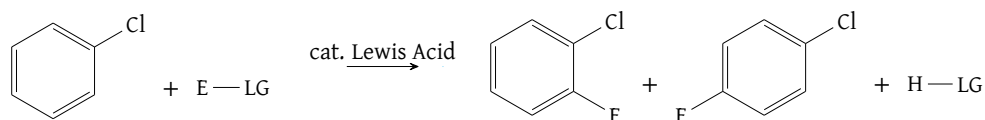


Figure 99: Halogen directing groups

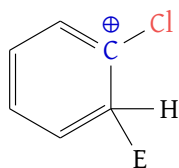
(a) Inductive Effect

Figure 100: Inductive effect

Under an O/P directing EAS interaction, inductive analysis would actually disfavor O/P substitution, since the resulting carbocation transition state places positive charge adjacent to the electronegative halogen, further depleting an already electron-poor center. The resulting dipole destabilizes the structure.

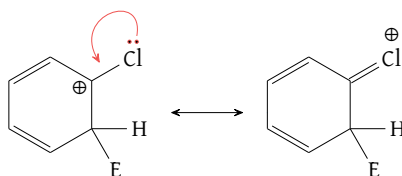
(b) Resonance

Figure 101: Resonance capability of halogen directing groups

Resonance, by contrast, allows the halogen to donate e^- density to satisfy the octet requirement throughout the molecule, favoring the O/P directing interaction.

Although these two effects act in opposition, experimental data confirm that resonance dominates over the inductive effect, resulting in O/P substitution being favored in practice.

3.2.3 Aromatic Substituent Manipulations

Core Concept 3.20: Reduction of Acyl Ketone

The reduction of an acyl ketone, previously employed in Core Concept 3.12, replaces a carbonyl oxygen with two hydrogen substituents.

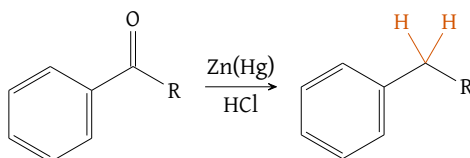


Figure 102: Reduction of an acyl ketone with Zn(Hg) and HCl

No mechanism is provided for this reaction.

Core Concept 3.21: Nitrobenzene Reduction to Aniline

The reduction of nitrobenzene proceeds analogously to the reduction of an acyl ketone (Core Concept 3.20), in which oxygens are replaced by hydrogens, and employs the same reactants to accomplish this transformation.

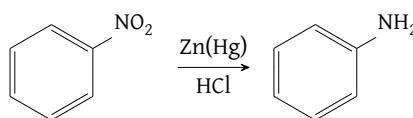


Figure 103: Nitrobenzene reduction to aniline with Zn(Hg) and HCl

No mechanism is provided for this reaction.

Core Concept 3.22: Diazotization of Aniline

The diazotization of aniline is a natural subsequent step following nitrobenzene reduction (Core Concept 3.21), as the product of the latter serves as the reactant for the former.

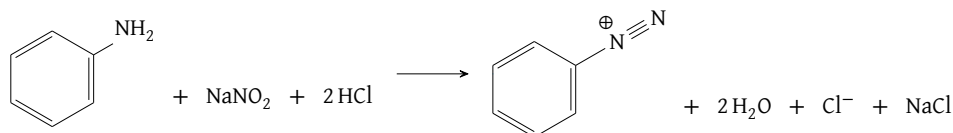
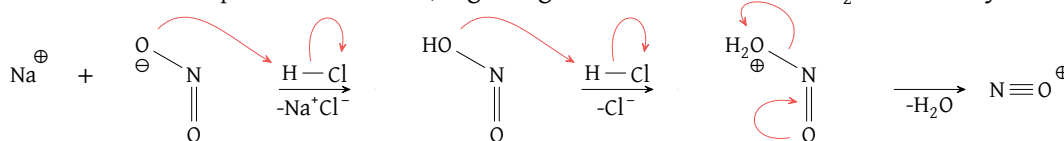


Figure 104: Diazotization of aniline

The reaction mechanism proceeds as follows, beginning with the conversion of NO_2^- to a nitrosyl cation.

Figure 105: Conversion of NO_2^- to a nitrosyl cation

The highly reactive nitrosyl cation then interacts with aniline as shown below to produce **aryl diazonium**.

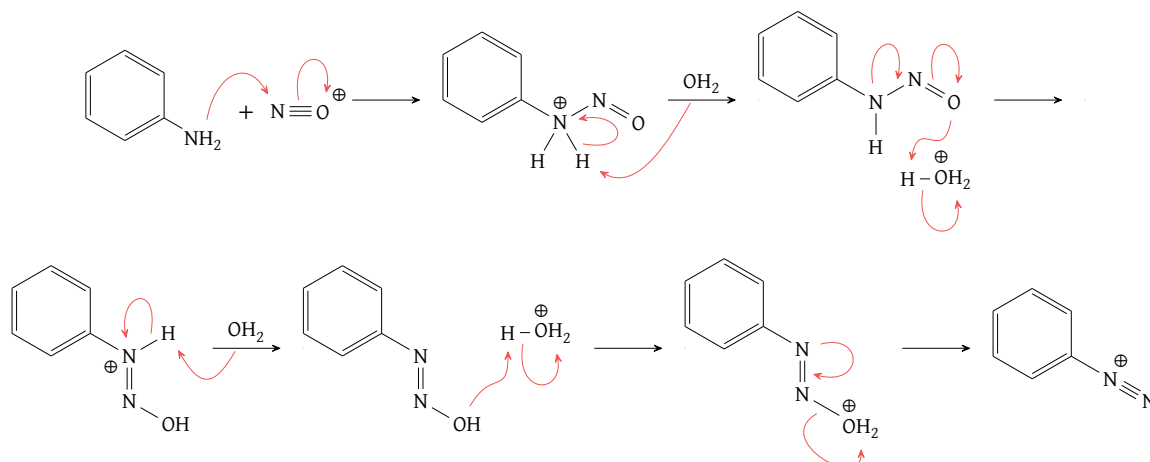


Figure 106: Conversion of aniline to aryl diazonium via interaction with the nitrosyl cation

3.3 Nucleophilic Aromatic Substitution

Core Concept 3.23: Nucleophilic Aromatic Substitution

Nucleophilic aromatic substitution (NAS or $\text{S}_{\text{N}}\text{Ar}$) is a reaction class in which the aromatic ring itself serves as the electrophile. NAS proceeds through one of two distinct pathways: **diazonium chemistry** (Core Concept 3.24) or **addition/elimination** (Core Concept 3.25).

Recall that in electrophilic aromatic substitution (Subsection 3.2), the aromatic ring instead serves as the nucleophile, the reverse of its role here.

Core Concept 3.24: Diazonium Chemistry

Diazonium chemistry begins when N_2 departs from a phenyldiazonium molecule via cleavage of the $\sigma \text{C}(\text{sp}^2)\text{-N}(\text{sp})$ bond, generating a benzene-ring-bearing carbocation alongside an exceptionally stable N_2 gas molecule.

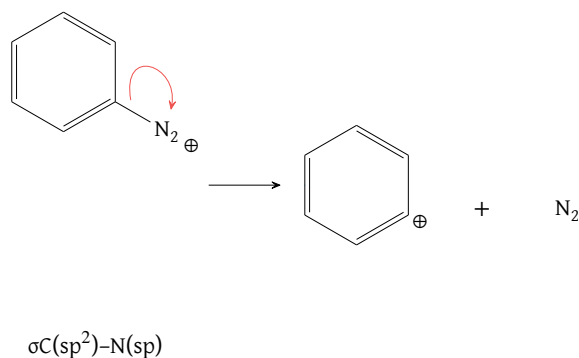


Figure 107: Diazonium chemistry; note that the empty orbital is the sp^2 orbital rather than the p orbital, meaning no resonance is available to stabilize the resulting positive charge

It is important to recognize that resonance stabilization is unavailable in this particular carbocation, since

the vacant orbital is the C(sp²) orbital rather than an unhybridized p orbital, which resonance would require. Consequently, this phenyl cation is highly reactive and readily seeks out any nearby nucleophile capable of stabilizing its positive charge.

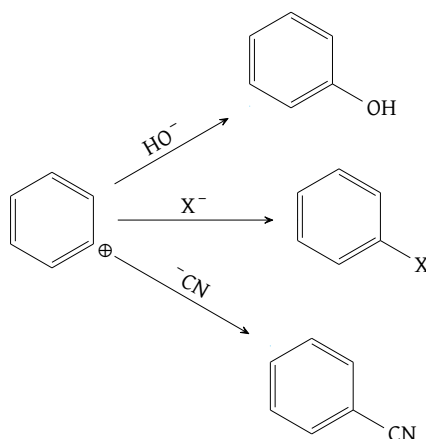


Figure 108: Nucleophilic attack on the benzene carbocation; most nucleophiles react readily here, with the notable exception of carbon-based nucleophiles such as Grignard reagents and organolithiums (Core Concept 2.3)

As illustrated above, nearly every nucleophile covered in this course participates readily in this reaction, with the exception of Grignard reagents and organolithiums. *The underlying reasoning for this exception falls outside the scope of these notes.*

By combining diazonium chemistry with reactions covered previously, chemists can effectively install nearly any nucleophile onto a benzene ring via the following reaction pathway.

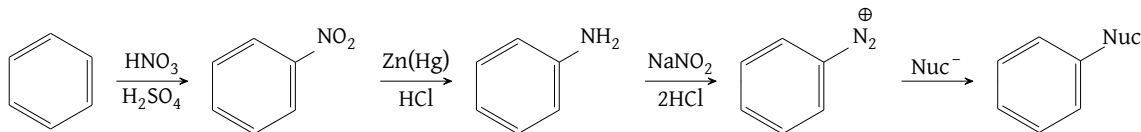


Figure 109: Complete reaction pathway illustrating nitration (Core Concept 3.10), nitro group reduction (Core Concept 3.21), diazotization (Core Concept 3.22), and diazonium chemistry (Core Concept 3.24)

Core Concept 3.25: Addition/Elimination

The NAS **addition/elimination** pathway proceeds according to the following general mechanism.

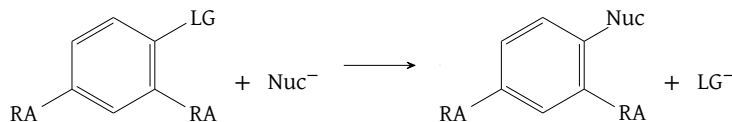


Figure 110: Outline of the nucleophilic aromatic substitution addition/elimination reaction; RA denotes a resonance acceptor, which must occupy the ortho or para position relative to the leaving group for the necessary resonance stabilization to occur

Recall that resonance acceptors were previously introduced in the context of meta-directing groups (Core Concept 3.18); here, however, their role is not to direct meta substitution but rather to accept electron density once the nucleophile attacks the carbon bearing the leaving group.

Additionally, this reaction **requires a strong nucleophile**, since aromaticity must ultimately be disrupted, as shown in the mechanism below. Note that this initial addition step is, consequently, the rate-determining step.

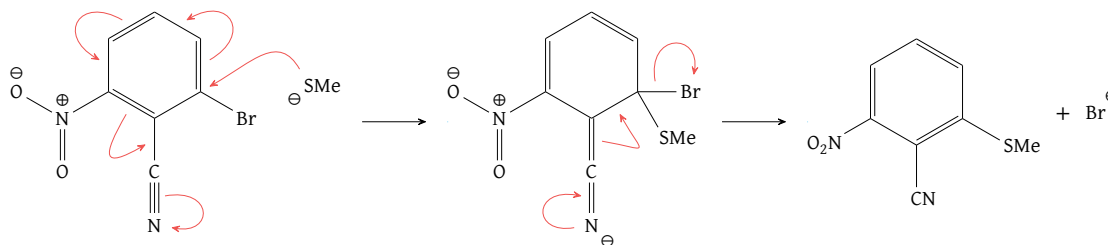


Figure 111: Nucleophilic aromatic substitution addition/elimination reaction; the NO_2 group cannot participate in resonance stabilization here, as it occupies the meta position

As the mechanism demonstrates, the reaction proceeds by relaying electron density into the resonance acceptor, which subsequently returns that electron density to expel the leaving group. Since the net outcome of this reaction resembles that of an $\text{S}_{\text{N}}2$ substitution, one might reasonably ask why an $\text{S}_{\text{N}}2$ -type pathway (Core Concept 2.2) is not viable here instead.

Revisiting the $\text{S}_{\text{N}}2$ mechanism, recall that, from a molecular orbital perspective, the nucleophile interacts with the σ^* anti-bonding orbital between the electrophile and leaving group, simultaneously expelling the leaving group and forming a new bond via backside attack.

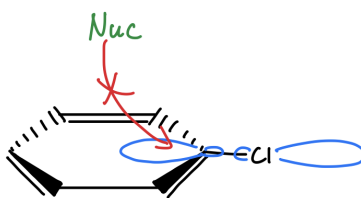


Figure 112: Backside nucleophilic attack on the carbon of chlorobenzene, prevented by steric hindrance from the aromatic ring

In the aromatic ring system, this necessary σ^* interaction is extremely unlikely, since the ring structure itself presents a substantial physical barrier to backside attack.

Within the framework of frontier molecular orbital theory, the π^* orbital lies lower in energy than σ^* , making it the LUMO and a far more accessible orbital for interaction. This is precisely what occurs in the addition/elimination pathway, which is consequently the favored mechanism.

Let us work through the following examples.

(a) Example 1: Predicting Major Kinetic Product

Consider the following reaction, which presents two possible leaving groups, F and Cl, alongside a NO_2 resonance acceptor.

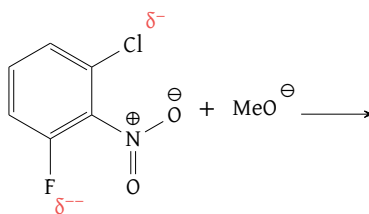


Figure 113: Reactants for NAS addition/elimination practice example 1

Both addition/elimination pathways are feasible, as shown below, ultimately leading to two distinct products.

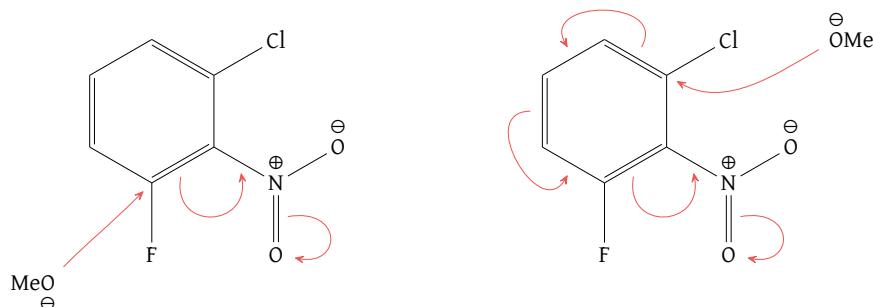


Figure 114: NAS addition/elimination practice example 1 — reaction pathways with fluorine (left) versus chlorine (right) serving as the leaving group

Because the nucleophile is held constant across both pathways, its ability to disrupt aromaticity is not the deciding factor. Instead, the outcome is governed by the larger dipole moment created by the more electronegative fluorine relative to chlorine. The resulting greater partial positive charge on the fluorine-bearing carbon renders it more electrophilic, so MeO^- preferentially attacks that position. The reaction therefore favors substitution at the carbon bearing F over the one bearing Cl.

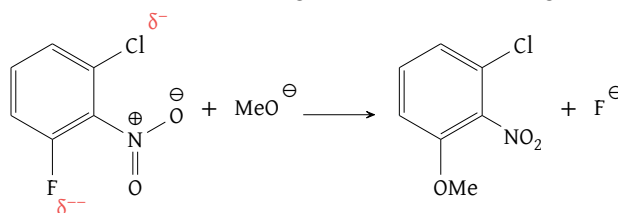


Figure 115: NAS addition/elimination practice example 1 — the correct reaction pathway

(b) Example 2: Predicting Product of NAS Chemistry

Consider the addition/elimination reaction below.

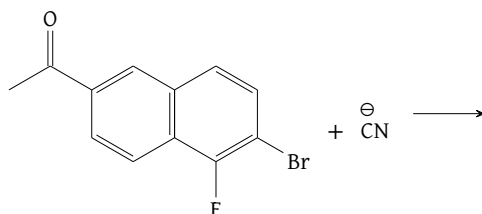


Figure 116: Reactants for NAS addition/elimination practice example 2

In this case, only one of the two possible leaving-group substitutions successfully donates e^- density into the resonance acceptor, while the other does not.

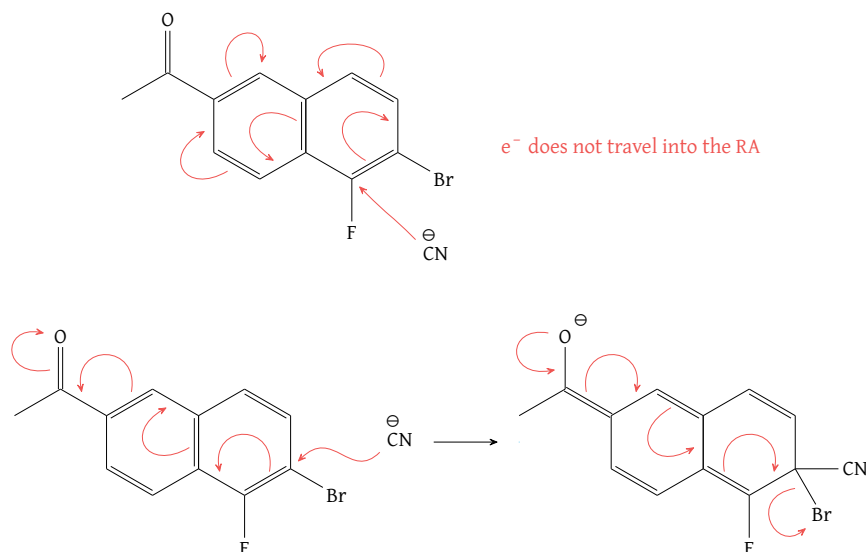


Figure 117: NAS addition/elimination practice example 2 — reaction pathways with fluorine (top) versus bromine (bottom) as the leaving group; note that treating fluorine as the leaving group prevents the resonance-accepting carbonyl group from accepting electron density, whereas treating bromine as the leaving group does not

The exclusive reaction outcome therefore proceeds as follows.

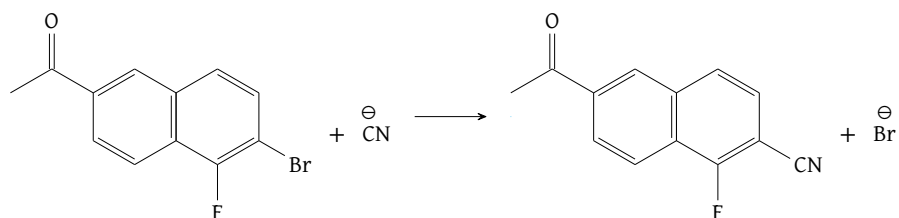


Figure 118: NAS addition/elimination practice example 2 — the correct reaction pathway

(c) Example 3: Predicting Starting Material

Consider the task of determining the starting aromatic compound that yields the following products, given the nucleophile shown.

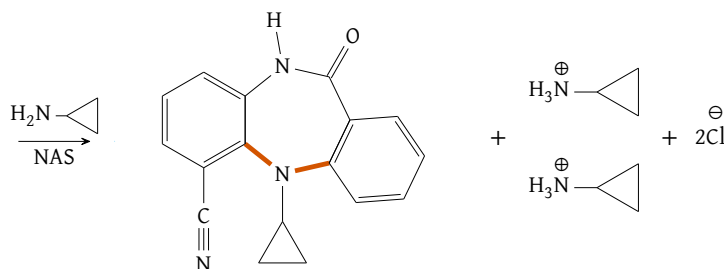


Figure 119: Products for NAS addition/elimination practice example 3

Because two Cl^- counterions and two $\text{H}_3\text{N}^+\text{R}$ groups appear on the product side, we can conclude that addition/elimination has occurred twice. The product conveniently shows the nucleophile positioned centrally, connected to the aromatic ring system through two bonds (orange) at the nitrogen. On this basis,

we can reasonably infer that the starting material bore leaving groups at precisely these two positions.

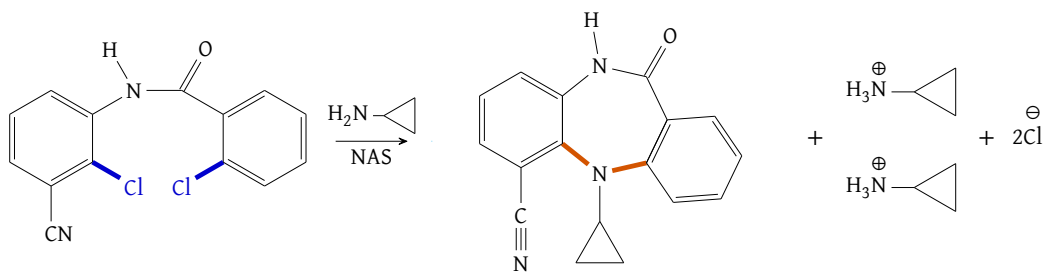


Figure 120: NAS addition/elimination practice example 3 — the complete reaction